

	FIR VEHICLE TESTING	14/07/2023
FIR No :FIR/Testing-Outdoor/517000/2023/6284		
Part No/ Group :51705156R		
Subject :Report on fluid economy evaluation of TML Signa 3521.T Cx 5L NGE (204 HP engine) OBD-II BS-VI vehicle with G950 (FGR: 9.36) Transmission and 5.58 Rear axle ratio in comparison with TML Signa 3521.T BS-VI OBD-I vehicle as a Base vehicle.		
Test Vehicle	Project Name :Signa 3521.T Cx 5L NGE (204 HP engine) Model Name : Signa 3521.T Cx 5L NGE (204 HP engine)	Enclosures : FIR
	Chasis Number :PC73550122RJ00814 Engine Number :GXX103612	
FIR Status :	Completed	

Role	Name	Signature
Prepared By	Debabrata Das	 14-Jul-2023
Checked By	Swapnil Jadhav	 14-Jul-2023
Approved By	Ajaykumar Jadhav	 17-Jul-2023

Cc: Head CVBU Engg, Chief Engineer, Head COC, Project Team [Design]; Project Team [Testing]

PA 20 - F 222 / May19

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Documents : Please click here

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 FIR/Testing-Outdoor/517000/2023/6284	FIR VEHICLE TESTING		Date: 13/07/2023	
Part No / Group: 51705156R	Specification no :- " 259051700014 "		Action by	Failure class

2.0 Result: Vide Observations		
FAILURES CLASS: A: Failure that happen without warning. Detrimental to vehicle / personnel, road users and safety. Can cause death / hospitalization. Noncompliance of regulatory requirement. B: Vehicle gets crippled with sufficient warning. Minor injury to occupant & road users (First aid only). Premature failures / repeated failures. Leads to consequential catastrophic damages. Need immediate attention. C: Customer irritants. Capable of damaging aesthetics. Failure can be attended at next servicing Schedule. D: Customer perception issues and minor irritants.		
3.0 Test Vehicles: - 1. Signa 3521.T BS-VI OBD-II (ERCJ-1259)_Test vehicle Chassis no. - PC73550122RJ00814; Engine no. - GXX103612. 2. Signa 3521.T BS-VI OBD-I (ERCJ 1164)_Base vehicle Chassis no. - PC53566012RJ00690; Engine no. - HZX503703.	Enclosures: Annexure 1: Panel chart Annexure 2: PAT sheet Annexure 3: Duty cycle report Annexure 4: Test request Annexure 5: Test plan	
Input Doc Ref.: - Test Request No: - Testing-Outdoor/15056 DID No: – NA. DDP No: – NA.		
Work Order No. :- P.22.EA.501	Make :- TATA Motors Limited	T I No. – 42503
Earlier Ref: - Nil.		
4.0 Project Status:- DR3		
5.0 Introduction: As a part of RDE (OBD-II) Testing Program, TML Signa 3521.T Cx 5L NGE BS-VI OBD-II vehicle with G950 (FGR:9.36) Transmission and 5.58 Rear axle ratio was offered for fluid economy evaluation along with TML Signa 3521.T 5L NGE BS-VI OBD-I vehicle as a base vehicle. All vehicles were tested back-to-back at JSR- Ranchi-JSR route in laden and unladen conditions at 40 km/hr as well as 55 km/hr vehicle cruising speeds. This report will reveal details of this trial with the status.		
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6.0 Testing:

Following test was conducted during evaluation :

Sr. no.	Test	Reference standard	Instruments used
1	Fluid economy evaluation	TST/VT/WG/158	Data Logger

7.0 Vehicle details:

Parameter		Base Vehicle	Test vehicle
Vehicle model		Signa 3521.T (OBD-I) (ERC-1164)	Signa 3521.T (OBD-II) (ERC-1259)
A) Vehicle specification			
Engine Model		Tata 5L NGE BS-VI (200 HP) OBD-I	Tata 5L NGE BS-VI (204 HP) OBD-II
Max Power		200 HP @2200 RPM	204 HP @2200 RPM
Max Torque		850 Nm @ 1000 ~1600 rpm	850 Nm @ 1000 ~1600 rpm
GVW (kg)		35000	35000
Gear Box		Model: Tata G950, Type: DD	Model: Tata G950, Type: DD
Gear Ratios.		1 st gear:9.36, 2 nd gear: 5.20, 3 rd gear: 3.05, 4 th gear: 1.84, 5 th gear: 1.33, 6 th gear: 1.00,	1 st gear:9.36, 2 nd gear: 5.20, 3 rd gear: 3.05, 4 th gear: 1.84, 5 th gear: 1.33, 6 th gear: 1.00,
Axle Ratio		5.12	5.58
Tires	FA1	Size: 295/90R20, Make: MRF , Type: non LCRR	Size: : 295/90R20, Make: Bridgestone, Type : LCRR (Phase-II)
	FA2	Size: 295/90R20, Make: MRF , Type: non LCRR	Size: : 295/90R20, Make: Bridgestone, Type : LCRR (Phase-II)
	LA1	Size: 295/90R20, Make: MRF , Type: non LCRR	Size: : 295/90R20, Make: Bridgestone, Type : LCRR (Phase-II)
	RA1	Size: 295/90R20, Make: MRF , Type: non LCRR	Size: : 295/90R20, Make: Bridgestone, Type : LCRR (Phase-II)
	RA2	Size: 295/90R20, Make: MRF , Type: non LCRR	Size: : 295/90R20, Make: Bridgestone, Type : LCRR (Phase-II)
Tractive effort @ 1 st gear		74851 N	81576 N
Tractive effort @ Top gear		7997 N	8715 N
Fan Details (Type/Make/Blades)		Type: E-Viscous fan and Size:24"	Type: E-Viscous fan and Size:24"



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Part No / Group: 51705156R		Specification no :- " 259051700014 "			Action by	Failure class

Parameter	Base Vehicle	Test vehicle
Vehicle model	Signa 3521.T (OBD-I) (ERC-1164)	Signa 3521.T (OBD-II) (ERC-1259)
B) FE Avenues _Engine Level		
Base Cal Improvement (Cal Maturity %)	As per production released	95% Cal Maturity
Low Viscous Engine Oil (TMGO-10W30)	No (15W40CK4+)	Yes (5W30)
FE Switch	Yes (3 Mode : E/B/P)	Yes (3 Mode : E/B/P)
Engine Features (EOL : Enable/Disable)	NA	NA
ATS Optimization	No	No
APM	No	No
Cruise Control	No	No
Gear based VAM	NA	NA
Intelligent mode shifter	NA	NA
Moving Mass Estimation	NA	NA
C) FE Avenues _Vehicle Level		
Driveline Optimization	NA	NA
GSA (Gear Shift Advisor)	Yes	Yes (Not Tuned)
MSA (Mode Shift Advisor)	NA	Yes (Not Tuned)
Low CRR Tyres	No	Yes
Low Viscous rear axle oil (80W90)	Yes	Yes
Smart Alternator (LIN)	NA	NA
THU/Unitised/Low friction bearing	Yes(Except R1)	Yes

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8.0 Test result:

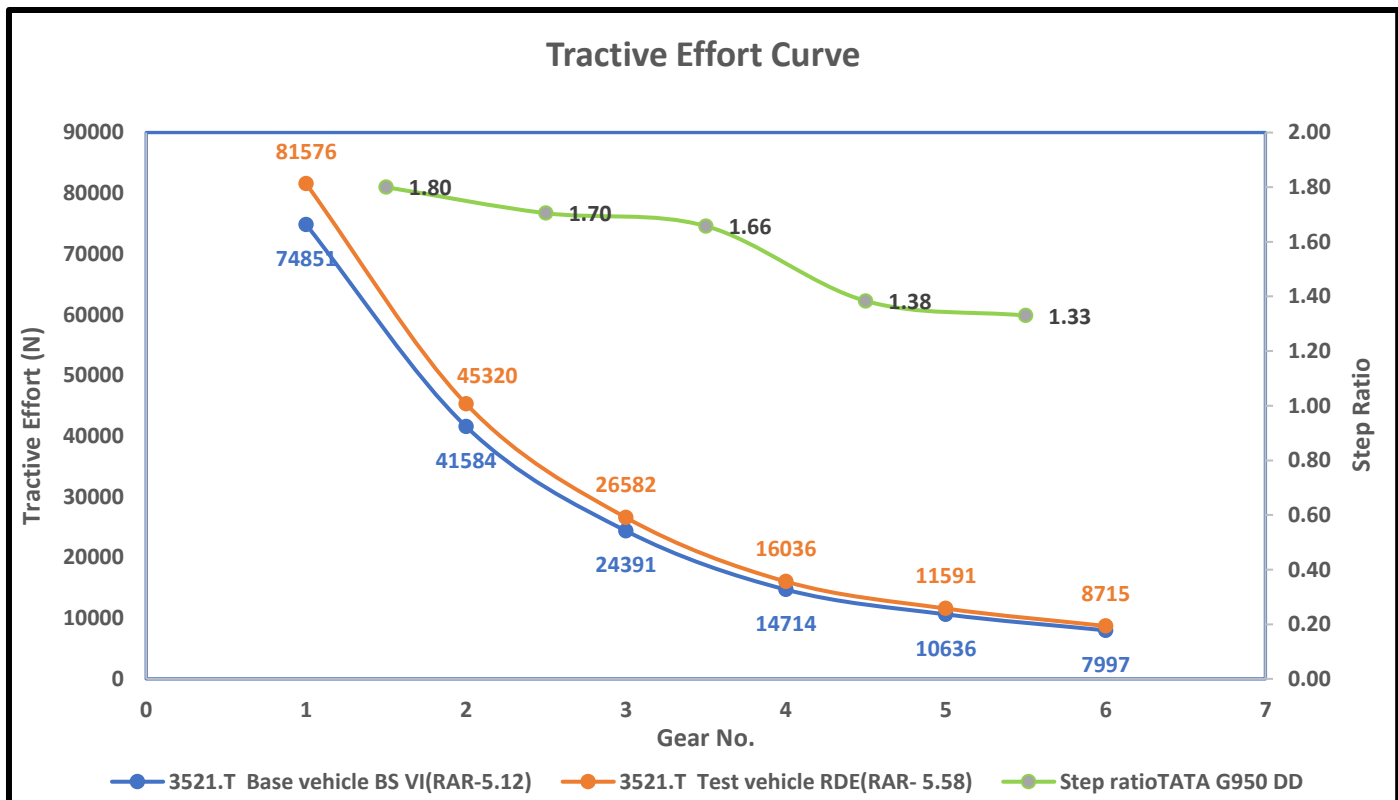
8.1 Cruising speed: 40 Kmph & 55 Kmph :

Parameters		Base Vehicle : QBD-1										Test Vehicle : QBD-2									
		SIGNA 3521.T Tata SL BSVI										SIGNA 3521.T Tata SL RDE BSVI									
1	Vehicle Identification No. (ERC No./FI No./Reg. No.)	ERC 1234										ERC 1239									
2	Vehicle GVW/GCVW	10000										10000									
3	Engine Details (Make/No./mm./cu. of Cy.)	TMS SLNGE										TMS SLNGE									
4	Engine Max Power	200 HP @ 2200 rpm										204 hp @ 2200 rpm									
5	Engine Max Torque	850 Nm @ 1500-1600 rpm										850 Nm @ 1500-1600 rpm									
6	Engine Calibration Maturity Level	As per production released										95%									
7	Regeneration Strategy (Time and/Seat base)	110 min										120 min									
8	Gear Ratios	TATA G-950 DD 1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00										TATA G-950 DD 1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00									
9	RAR	5.12										5.58									
10	Type Details	295/90 R 20 MB MRF run low LCRB										295/90R 20 MB Bridgestone Low LCRB									
11	LBCB	High & Low Engine Speed Breakpoint																			
	Vehicle Mass Threshold	kg										--									
	Variable Acceleration 1, 2	kmph/Vac										--									
	Variable Acceleration Speed 1, 2	kmph										--									
	GD Max. Veh. Speed (Heavy & Light Engine Load)	kmph										--									
	RSG	Maximum Accelerator Vehicle Speed										80									
	Torque Limit Medium Adjustment Factor	Balance										100%									
12	Twist	Torque Limit Maximum Adjustment Factor										100%									
		Powertrain Protection (Enable / Disable)										Disable									
	PTP	Torque Limit										Nm									
13	USE / Torque Table Revision (1/2/3/N)	NA										NA									
14	Fun Details (Type/Make/Model)	E- 34" High Efficiency Vortexon Fan										24" High Efficiency Vortexon Fan									
	Base Cal Improvement (Cal Maturity %)	As per production released										95% Cal Maturity									
	Low Viscous Engine Oil (TMGO-SW30)	Yes (SW400X4)										Yes (SW400X4)									
	FE Switch	Yes (3 Mode: 6/8/FP)										Yes (3 Mode: 6/8/FP)									
	Engine Features (EOC : Enable/Disable)	Disable										No									
15	ATS Optimization	No										No									
	APM	No										No									
	Cruise Control	No										No									
	Gear based VMM	No										No									
	Intelligent mode shifter	No										No									
	Moving Mass Estimation	No										No									
	Driveline Optimization	No										No									
	GSA (Gear Shift Actuator)	Yes										Yes (No Torque)									
	MSA (Mode Shift Actuator)	No										No									
	Low CRB Types	No										Yes									
16	Low Viscous rear axle oil (BW90)	Yes										Yes									
	Smart Absorber (L&R)	No										No									
	THU(Disabled)/Low Friction bearing	Yes (Exempt RS)										No									
	Route	Jamshedpur -Ranchi -Jamshedpur										Jamshedpur -Ranchi -Jamshedpur									
	FE Duty Cycle (Laden/Unladen)	Laden Trial @ 40 kmph										Laden Trial @ 55 kmph									
17	FE Trial Details	25.04.2023										25.04.2023									
	Weather Condition (Sunny/Rainy/Cloudy/Windy)	Sunny										Sunny									
	Traffic Condition (Light/Heavy/Heavy)	Sunny										Sunny									
	Regeneration (Off)	Average										Average									
	Fuel spent in Regen	%										%									
	Fuel spent in SCRTM	%										%									
	Trips Distance	km										km									
	Trips Fuel Quantity	Litre										Litre									
	Total Urea Quantity	Litre										Litre									
	Equivalent Urea Quantity : 50% Consumption	Litre										Litre									
	Total Eq. Fluid Consumed	Litre										Litre									
	Urea loss as % of Total	%										%									
	Fuel Economy	kmpl										kmpl									
	Status of Fuel economy	%										%									
	Equivalent Fuel Economy	kmpl										kmpl									
	Status of Equivalent Fuel Economy	%										%									
	Cluster Total Ods. Reading (at end of trial)	km										km									
	Cluster Trip AFE	kmpl										kmpl									
	Status of Cluster Trip AFE with Actual Fuel economy	%										%									
	18	Average Fuel Economy	kmpl										kmpl								
Status of Average Fuel economy		%										%									
Average Eq. Fluid Economy		kmpl										kmpl									
Status of Average Eq. Fluid economy		%										%									
Overall Avg. Fuel Economy (kmpl) : L + UL		%										%									
Status of Overall Avg. Fuel Economy (%)	%										%										
19	Overall Avg. Eq. Fluid Economy (kmpl) : L + UL	%										%									
	Status of Overall Avg. Eq. Fluid Economy (%)	%										%									
	Overall Avg. Fuel Economy (kmpl) : L + UL	%										%									
	Status of Overall Avg. Fuel Economy (%)	%										%									
	Overall Avg. Eq. Fluid Economy (kmpl) : L + UL	%										%									
20	Status of Overall Avg. Eq. Fluid Economy (%)	%										%									
	Average Fuel Economy	kmpl										kmpl									
	Status of Average Fuel economy	%										%									
	Average Eq. Fluid Economy	kmpl										kmpl									
	Status of Average Eq. Fluid economy	%										%									
21	Overall Avg. Fuel Economy (kmpl) : L + UL	%										%									
	Status of Overall Avg. Fuel Economy (%)	%										%									
	Overall Avg. Eq. Fluid Economy (kmpl) : L + UL	%										%									
	Status of Overall Avg. Eq. Fluid Economy (%)	%										%									
	Overall Avg. Fuel Economy (kmpl) : L + UL	%										%									

Note: Equivalent Urea consumption is taken as 50% of actual urea consumption.

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8.3 Tractive Effort Comparison :



Remarks:

Signa 3521.T Cx BS-VI OBD-II (204 HP) Test vehicle tractive effort is higher in all gears as compared with Signa 3521.T BS-VI OBD-I (200 HP) Base vehicle.

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8.4 Torque Table :

8.3.1 : Base vehicle (Signa 3521.T BS-VI OBD-I (200 HP))

FE Switch Torque Table (Torque Limit)										
	Load Gea	1st Gear	2nd Gear	3rd Gear	4th Gear	5th Gear	6th Gear	7th Gear	8th Gear	Reverse
Economy Mode	-	850	850	700	590	590	590	-	-	850
Balance Mode	-	850	850	700	700	700	700	-	-	850
Power Mode	-	850	850	850	850	850	850	-	-	850

8.3.2: Test vehicle (Signa 3521.T BS-VI OBD-II (204 HP))

FE Switch Torque Table (Torque Limit)										
	Load Gea	1st Gear	2nd Gear	3rd Gear	4th Gear	5th Gear	6th Gear	7th Gear	8th Gear	Reverse
Economy Mode	NA	850	850	700	590	590	590	-	-	850
Balance Mode	NA	850	850	700	700	700	700	-	-	850
Power Mode	NA	850	850	850	850	850	850	-	-	850

9.0 Observation: Nil

10.0 Further Testing: Further fuel economy evaluation needs to be done with benchmark vehicle to get the PALS attribute wrt core competitor.

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Level 1 Attribute:		PANEL Chart		VE CVBU/FE/01
PAT- Fuel Economy		Specification:Max Power- 204 hp @ 2200 rpm Max Torque - 850 Nm 1000-1600 rpm Driveline : G950 DD 5.58 RAR 295/90R 20 Tyres 3 Mode FE Switch Torque Factor: NA EOL Parameters : RSG: 80 kmph		Panel Chart No: VE CVBU/ HCV/ SIGNA 3521.T 5L Cx BSVI RDE/ Cx / FE/01
Programme / MY : SIGNA 3521.T Cx 5L BSVI RDE	Gateway : DR3	Owner: Ajaykumar Jadhav	Project Name: SIGNA 3521.T 5L BSVI RDE(Cx Version)	Date: 12/07/2023

Level 1 Summary:			Statement Of Intent (SOI) or DNA Description:	
Level 1:	PALS Position	Status		Fuel Economy: Haulage Application
		Current/ Gateway	Job#1 Predicted	
	L	L		

Attribute Target Summary:					Status				Base Vehicle		Remarks
Level 2	Level 3	Level 4	Level 5	Level 6	PALS	Metric	Target	Current/Gateway SIGNA 3521.T Cx BSVI RDE	Job#1 Predicted	Base Signa 3521.T BSVI	
Specification								With THU bearing	Without THU bearing Extrapolated	--	
Vehicle GVW (Payload)						kg		35000		35000	
Power (Max) @ rpm						HP		204 hp @ 2200 rpm		200 hp @ 2200 rpm	
Power to Weight ratio						HP/Ton		5.83		5.71	
Torque (Max) @ rpm						Nm		850 Nm @ 1000-1600 rpm,		850 Nm @ 1000-1600 rpm	
Torque to Weight ratio						Nm/Ton		24.29		24.29	
Tyres size						--		Bridgestone Tyre LCCR 295/90R20		MRF Tyre non LCCR 295/90R20	
Gear Box Ratios						--		G950 DD, FGR -9.36		G950 DD, FGR -9.36	
Final Drive Ratio						--		5.58		5.12	
Fluid Economy at Rated Load					L			L	L		
Highway AC OFF						--					
Route : Jamshedpur-Ranchi						--					
Avg. Fluid Economy @40 kmph						kmpl	≥ 3.68	3.78	3.76	3.68	*Signa 3521.T OBII Cx and Signa 3521.T OBDI was tested back to back condition on JSR-Ranchi route.
Avg. Fluid Economy @55 kmph						kmpl	≥ 3.37	3.51	3.49	3.37	
Fluid Economy At Unladen					L			L	L		
Highway AC OFF						--					
Route : Jamshedpur-Ranchi						--					
Avg. Fluid Economy @40 kmph						kmpl	≥ 7.08	7.21	7.17	7.08	*Signa 3521.T OBII Cx and Signa 3521.T OBDI was tested back to back condition on JSR-Ranchi route.
Avg. Fluid Economy @55 kmph						kmpl	≥ 6.21	6.38	6.35	6.21	
Overall Fuel Economy (80% Laden + 20% Unladen)					L			L	L		
Highway AC OFF						--					
Route : Jamshedpur-Ranchi						--					
Overall Avg. Fluid Economy @40 kmph						kmpl	≥ 4.07	4.17	4.15	4.07	*Signa 3521.T OBII Cx and Signa 3521.T OBDI was tested back to back condition on JSR-Ranchi route.
Overall Avg. Fluid Economy @55 kmph						kmpl	≥ 3.70	3.86	3.84	3.70	

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 TATA MOTORS LTD. ERC,PUNE		PAT ATTRIBUTES SHEET - Fuel Economy										VE CVBU/FE/01							
		Depart ment Name: Vehicle Engineering					Project Name: SIGNA 3521.T 5L BSVI RDE Cx (G-950 DD FGR-9.36, 5.58 RAR, 295/90R 20 phase II low CRR Tyres, 3 Mode FE Switch) Base Vehicle: SIGNA 3521.T 5L BSVI OBD-1, G-950 DD, FGR-9.36, 5.12 RAR, 295/90R 20 Tyres, 3 Mode FE Switch Application: Haulage					PAT Sheet No.: VE CVBU/ HCV/ SIGNA 3521.T 5L BSVI RDE/ Cx / FE/01							
												Date: 12/07/2023							
Sr.No.	Aggregate and test parameter						Unit	Test Type	Classification @OR4	Ref Standard / Work Guideline	Simulation	Base Vehicle SIGNA 3521.T BSVI OBD-1 (G 950 DD RAR 5.12 295/90R20)	Target	Test Vehicle SIGNA 3521.T BSVI OBD-II Cx (G 950 DD RAR 5.58 295/90R20)		Status (RYG)	Remarks	CoC Remark	
	Level 2	Level 3	Level 4	Level 5	Level 6	Level 7													
A	Fuel Economy Evaluation										Heading		--		--		--		
2.0	Fuel Economy At Rated Load										Heading								
2.7	Highway AC OFF										Heading				with THU Bearing	without THU Bearing			
2.7.1	Route : Jamshedpur-Ranchi								Specification	Standard	TSTVTWQ158								
2.7.9	Average Fuel Economy @40 kmph						Kmpl		Data Generation				3.77	--	3.86	3.84	--	1. Test vehicle SIGNA 3521.T Cx BSVI OBD-2 tested with : 204 HP Engine calibration with 95% Cal Maturity + THU Bearing + Low viscous 5W/30 engine oil + Low viscous 80W90 pro FE rear axle oil + Bridgestone LCRR tyres + 24" Fan 2. Target has been taken at par (As per Cx approach) wrt to base SIGNA 3521.T OBD-1 BSVI vehicle. 3. SIGNA 3521.T Cx RDE BSVI is meeting the PALS target of "Leadership". 4. Concluded fuel economy results is extrapolated for Cx variant to reducing the THU bearing contribution by 0.5% from actual fuel economy results.	
2.7.10	Average Fuel Economy @55 kmph						Kmpl		Data Generation				3.43	--	3.58	3.56	--		
2.7.11	Average Fuel Economy @40 kmph						Kmpl		Target				3.68	≥ 3.68	3.78	3.76	G		
2.7.12	Average Fuel Economy @55 kmph						Kmpl		Data Generation				3.37	≥ 3.37	3.51	3.49	G		
2.8	Fuel Economy At Unladen										Heading								
4.7	Highway AC OFF										Heading								
4.7.1	Route : Jamshedpur-Ranchi								Specification	Standard	TSTVTWQ158								
4.7.9	Average Fuel Economy @40 kmph						Kmpl		Data Generation				7.31	--	7.45	7.41	--	1. Test vehicle SIGNA 3521.T Cx BSVI OBD-2 tested with : 204 HP Engine calibration with 95% Cal Maturity + THU Bearing + Low viscous 5W/30 engine oil + Low viscous 80W90 pro FE rear axle oil + Bridgestone LCRR tyres + 24" Fan 2. Target has been taken at par (As per Cx approach) wrt to base SIGNA 3521.T OBD-1 BSVI vehicle. 3. SIGNA 3521.T Cx RDE BSVI is meeting the PALS target of "Leadership". 4. Concluded fuel economy results is extrapolated for Cx variant to reducing the THU bearing contribution by 0.5% from actual fuel economy results.	
4.7.10	Average Fuel Economy @55 kmph						Kmpl		Data Generation				6.36	--	6.47	6.44	--		
4.7.11	Average Fuel Economy @40 kmph						Kmpl		Target				7.08	≥ 7.08	7.21	7.17	G		
4.7.12	Average Fuel Economy @55 kmph						Kmpl		Data Generation				6.21	≥ 6.21	6.38	6.35	G		
5.0	Overall Fuel Economy (80% Laden + 20% Unladen)										Heading								
5.1	Highway AC OFF										Heading								
5.1.1	Route : Jamshedpur-Ranchi								Specification	Standard	TSTVTWQ158								
5.1.9	Average Overall Fuel Economy @40 kmph						Kmpl		Data Generation				4.17	--	4.27	4.25	--	1. Test vehicle SIGNA 3521.T Cx BSVI OBD-2 tested with : 204 HP Engine calibration with 95% Cal Maturity + THU Bearing + Low viscous 5W/30 engine oil + Low viscous 80W90 pro FE rear axle oil + Bridgestone LCRR tyres + 24" Fan 2. Target has been taken at par (As per Cx approach) wrt to base SIGNA 3521.T OBD-1 BSVI vehicle. 3. SIGNA 3521.T Cx RDE BSVI is meeting the PALS target of "Leadership". 4. Concluded fuel economy results is extrapolated for Cx variant to reducing the THU bearing contribution by 0.5% from actual fuel economy results.	
5.1.10	Average Overall Fuel Economy @55 kmph						Kmpl		Data Generation				3.78	--	3.93	3.91	--		
5.1.11	Overall Average Fuel Economy @40 kmph						Kmpl		Target				4.07	≥ 4.07	4.17	4.15	G		
5.1.12	Overall Average Fuel Economy @55 kmph						Kmpl		Data Generation				3.79	≥ 3.79	3.86	3.84	G		
Sign Off												Index for Status							
Responsibility		Name		Signature		Off target, no resolution													
PAT Engineer		Palash Das /Debabrata Das/ Swapnil Jadhav		  		Marginally off target, resolution planned/low risk													
PAT Lead / Project Lead		Ajaykumar Jadhav / Sanjeev kumar		 		Met target, no concern													

OUR CULTURE CREDO

AT TATA MOTORS

We are connecting aspirations by being bold in thought and action, owning every opportunity and challenge, Solving together as one team and engaging all our stakeholders with empathy.

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Signa 3521.T RDE Vehicle Fuel Economy Evaluation status with FE Avenues

PED - HCV | 11.07.2023



Sr. No.	Parameters		Base Vehicle : OBD-1	Test Vehicle : OBD-2	
			SIGNA 3521.T Tata 5L BSVI	SIGNA 3521.T Tata 5L RDE BSVI	
1	Vehicle Identification No. (ERC No. /EJ No./Reg. No.)		ERC-1164	ERC-1259	
2	Vehicle GVW/GCW	kg	35000	35000	
3	Engine Details (Make/cc/Norms/no. of Cyl.)		TML 5LNGE	TML 5LNGE	
	Engine Max Power		200 HP @ 2200 rpm	204 hp @ 2200 rpm	
	Engine Max Torque		850 Nm @ 1000-1600 rpm	850 Nm @ 1000-1600 rpm	
	Engine Calibration Maturity Level		As per production released	95%	
4	Regeneration Strategy (Time ase/Soot base)		110 Hrs	110 Hrs	
5	Gear Box (make/drive/FGR)		TATA G-950 DD	TATA G-950 DD	
	Gear Ratios		1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00	1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00	
6	RAR		5.12	5.58	
7	Tyre Details		295/90 R 20 RIB MRF low LCRR	295/90R 20 RIB Bridgestone Low LCRR	
8	LBSC	High & Low Engine Speed Breakpoint	RPM	--	
		Vehicle Mass Threshold	kg	--	
	VAM	Variable Acceleration 1, 2	kmph/sec	--	
		Variable Acceleration Speed 1, 2	kmph	--	
	GDP	GD Max. Veh. Speed (Heavy & Light Engine Load)	kmph	--	
	RSG	Maximum Accelerator Vehicle Speed	kmph	80	
	Torque Factor	Torque Limit Medium Adjustment Factor	Balance	100%	--
		Torque Limit Maximum Adjustment Factor	Economy	100%	--
	PTP	Powertrain Protection (Enable / Disable)		Disable	Disable
		Torque Limit	Nm	NA	NA
USFE / Torque Table Revision (1/2/3/n)			--	--	
9	Fan Details (type/make/blades)		E- 24" High Efficiency Viscous fan	24" High Efficiency Viscous fan	

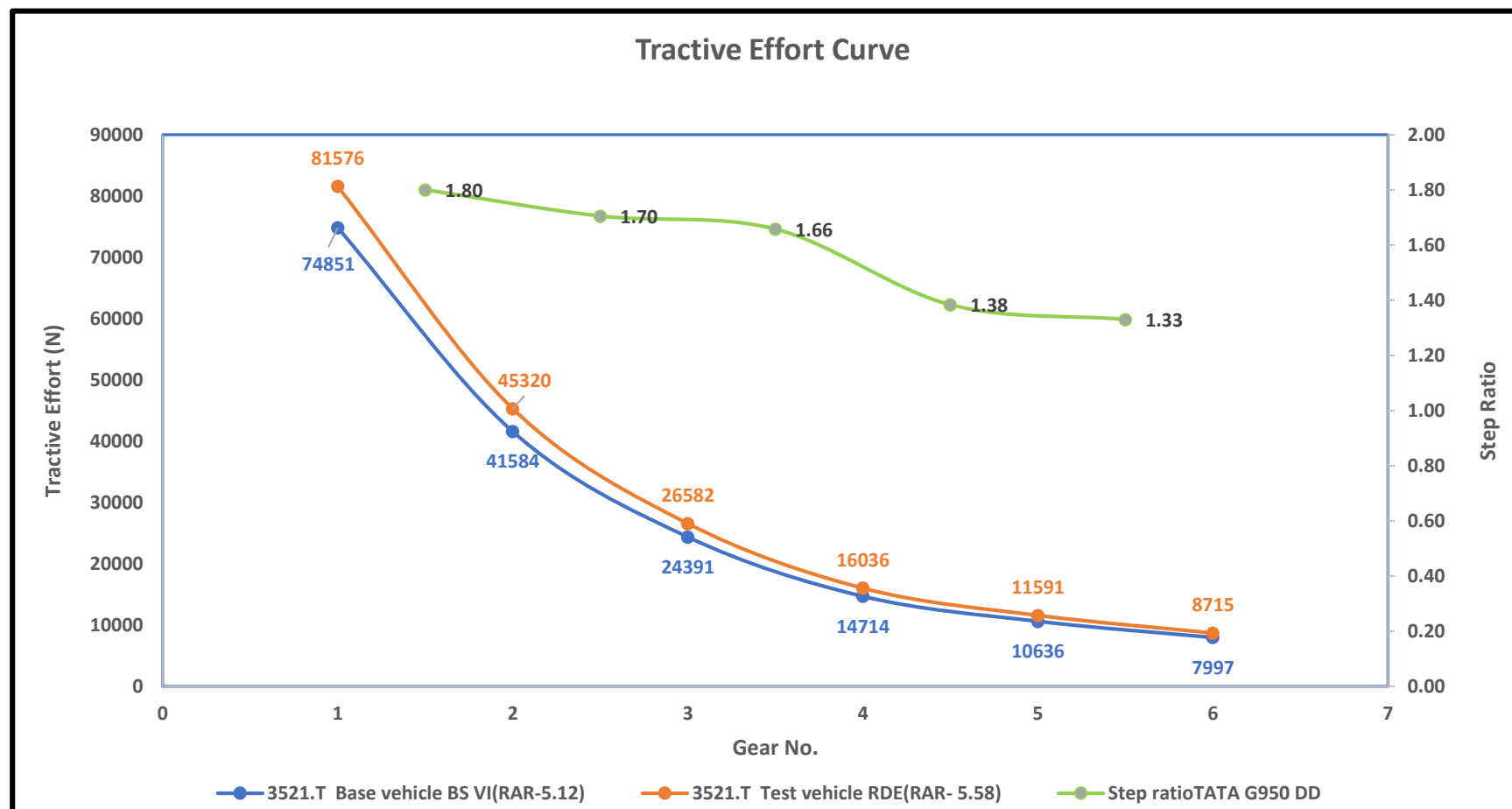
FE Avenues

Sr. No.	Parameters	Base Vehicle : OBD-1	Test Vehicle : OBD-2
		SIGNA 3521.T Tata 5L BSVI	SIGNA 3521.T Tata 5L RDE BSVI
FE Avenues (Engine Level)	Base Cal Improvement (Cal Maturity %)	As per production released	95% Cal Maturity
	Low Viscous Engine Oil (TMGO-10W30)	No (15W40CK4+)	Yes (15W40CK4+)
	FE Switch	Yes (3 Mode : E/B/P)	Yes (3 Mode : E/B/P)
	Engine Features (EOL : Enable/Disable)	Disable	Disable
	ATS Optimization	No	No
	APM	No	No
	Cruise Control	No	No
	Gear based VAM	No	No
	Intelligent mode shifter	No	No
FE Avenues (Vehicle Level)	Moving Mass Estimation	No	No
	Driveline Optimization	NA	No
	GSA (Gear Shift Advisor)	Yes	Yes (Not Tuned)
	MSA (Mode Shift Advisor)	NA	No
	Low CRR Tyres	No	Yes
	Low Viscous rear axle oil (80W90)	Yes	Yes
	Smart Alternator (LIN)	NA	No
	THU/Unitised/Low friction bearing	Yes (Except R1)	Yes

FE Summary

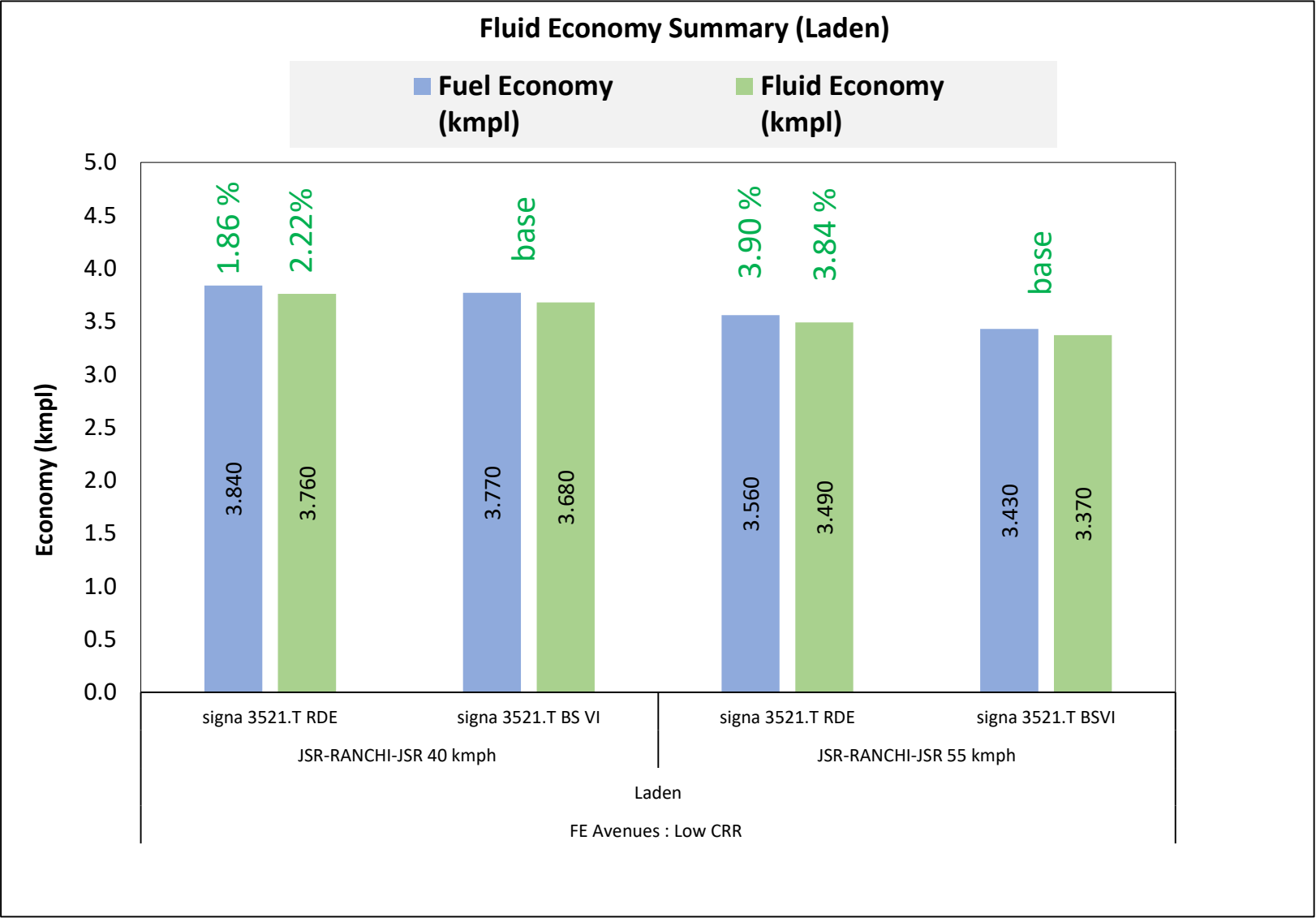
Sr. No.	Parameters		Base Vehicle : OBD-1										Test Vehicle : OBD-2												
			SIGNA 3521.T Tata 5L BSVI										SIGNA 3521.T Tata 5L RDE BSVI												
FE Trial Details	Route		Jamshedpur-Ranchi-Jamshedpur										Jamshedpur-Ranchi-Jamshedpur												
	FE Duty Cycle (Laden/Unladen)		Laden Trial @ 40 kmph			Unladen @ 40 kmph		Laden Trial @ 55 Kmph			Unladen @ 55 kmph		Laden Trial @ 40 kmph			Unladen @ 40 kmph		Laden Trial @ 55 Kmph			Unladen @ 55 kmph				
	Trial Date		25.04.2023	26.04.2023	27.04.2023	02.05.2023	03.05.2023	20.04.2023	22.04.2023	23.04.2023	24.04.2023	05.05.2023	06.05.2023	25.04.2023	26.04.2023	27.04.2023	02.05.2023	03.05.2023	20.04.2023	22.04.2023	23.04.2023	24.04.2023	05.05.2023	06.05.2023	
	Weather Condition (Sunny/Rainy/Cloudy/Windy)		Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	
	Traffic Condition (Light/Average/Heavy)		Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	
	Regeneration (Yes/No)		No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	
	Fuel spent in Regen		%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	
	Fuel spent in SCRTM		%	Data Awaited	Data Awaited	Data Awaited	Data Awaited	1.85%	Data Awaited	Data Awaited	Data Awaited	Data Awaited	Data Awaited	Data Awaited	Data Awaited	Data Awaited	Data Awaited	Data Awaited	Data Awaited	Data Awaited	Data Awaited	Data Awaited	Data Awaited	Data Awaited	
	Trip Distance		km	241.00	241.00	241.00	220.00	241.00	202.00	241.00	241.00	241.00	150.00	241.00	241.00	241.00	220.00	241.00	202.00	241.00	241.00	241.00	150.00	241.00	
	Total Diesel Quantity		Litre	61.45	64.69	63.28	28.56	34.85	49.67	69.69	70.05	71.24	23.65	37.76	63.18	62.36	62.63	28.19	33.92	49.94	67.74	66.36	67.95	23.28	37.05
	Total Urea Quantity		Litre	3.80	4.20	2.20	2.70	1.20	2.40	2.40	2.40	3.00	1.30	1.70	2.20	3.30	2.00	3.00	0.90	1.00	3.00	2.30	2.40	0.60	1.20
	Equivalent Urea Quantity : 50% Consumption		Litre	1.90	2.10	1.10	1.35	0.60	1.20	1.20	1.20	1.50	0.65	0.85	1.10	1.65	1.00	1.50	0.45	0.50	1.50	1.15	1.20	0.30	0.60
	Total Eq. Fluid Consumed		Litre	63.35	66.79	64.38	29.91	35.45	50.87	70.89	71.25	72.74	24.30	38.61	64.28	64.01	63.63	29.69	34.37	50.44	69.24	67.51	69.15	23.58	37.65
	Urea uses % wrt. Diesel		%	6%	6%	3%	9%	3%	5%	3%	3%	4%	5%	8%	3%	5%	3%	11%	3%	2%	4%	3%	4%	3%	3%
	Fuel Economy		kmpl	3.92	3.73	3.81	7.70	6.92	4.07	3.46	3.44	3.38	6.34	6.38	3.81	3.86	3.85	7.80	7.10	4.04	3.56	3.63	3.55	6.44	6.50
	Status of Fuel economy		%	Base	Base	Base	Base	Base	Base	Base	Base	Base	Base	Base	-2.74%	3.74%	1.04%	1.31%	2.74%	-0.54%	2.88%	5.56%	4.84%	1.59%	1.92%
	Equivalent Fluid Economy		kmpl	3.80	3.61	3.74	7.36	6.80	3.97	3.40	3.38	3.31	6.17	6.24	3.75	3.77	3.79	7.41	7.01	4.00	3.48	3.57	3.49	6.36	6.40
	Status of Equivalent Fluid economy		%	Base	Base	Base	Base	Base	Base	Base	Base	Base	Base	Base	-1.4%	4.3%	1.18%	0.7%	3.1%	0.85%	2.38%	5.54%	5.19%	3.05%	2.55%
	Cluster Total Odo. Reading (at End of trial)		km	245	245	245	231	251	213	254	254	254	155	251	240	240	218	238	201	240	240	240	146	237	
	Cluster Trip AFE		kmpl	4.40	4.50	4.40	8.00	8.00	3.80	3.50	3.50	3.50	6.50	6.50	4.10	4.10	4.10	8.00	7.80	4.20	3.80	3.80	3.70	6.50	7.00
Status of Cluster Trip AFE wrt Actual fluid economy		%	15.66%	20.79%	15.53%	3.85%	15.68%	-6.56%	1.21%	1.73%	3.46%	2.48%	1.84%	7.48%	6.09%	6.55%	2.51%	9.78%	3.84%	6.81%	4.63%	4.32%	0.88%	7.61%	
Avg.kmpl Results	Average Fuel Economy		kmpl	3.77			7.31			3.43			6.36		3.86			7.45		3.58			6.47		
	Status of Average Fuel economy		%	Base			Base			Base			Base		2.37%			1.99%		4.42%			1.75%		
	Average Eq. Fluid Economy		kmpl	3.68			7.08			3.37			6.21		3.78			7.21		3.51			6.38		
	Status of Average Eq. Fluid economy		%	Base			Base			Base			Base		2.73%			1.89%		4.36%			2.80%		
Overall Results L+UL	Overall Avg. Fuel Economy (kmpl) : L + UL		80%	20%	4.17			3.78			4.27			3.93											
	Status of Overall Avg. Fuel Economy (%)					Base			Base			2.33%			4.10%										
	Overall Avg. Eq. Fluid Economy (kmpl) : L + UL		80%	20%	4.07			3.70			4.17			3.86											
	Status of Overall Avg. Eq. Fluid Economy (%)					Base			Base			2.63%			4.17%										
THU Bearing is not present in Live Axle. So,THU contribution is deducted (0.25 per hub *2=0.5) from Test Vehicle FE & FLE value																									
Avg.kmpl Results	Average Fuel Economy		kmpl	3.77			7.31			3.43			6.36		3.84			7.42		3.56			6.44		
	Status of Average Fuel economy		%	Base			Base			Base			Base		1.86%			1.48%		3.90%			1.24%		
	Average Eq. Fluid Economy		kmpl	3.68			7.08			3.37			6.21		3.76			7.17		3.49			6.35		
	Status of Average Eq. Fluid economy		%	Base			Base			Base			Base		2.22%			1.38%		3.84%			2.29%		
Overall Results L+UL	Overall Avg. Fuel Economy (kmpl) : L + UL		80%	20%	4.17			3.78			4.25			3.91											
	Status of Overall Avg. Fuel Economy (%)					Base			Base			1.82%			3.58%										
	Overall Avg. Eq. Fluid Economy (kmpl) : L + UL		80%	20%	4.07			3.70			4.15			3.84											
	Status of Overall Avg. Eq. Fluid Economy (%)					Base			Base			2.12%			3.65%										

Tractive Effort Comparison

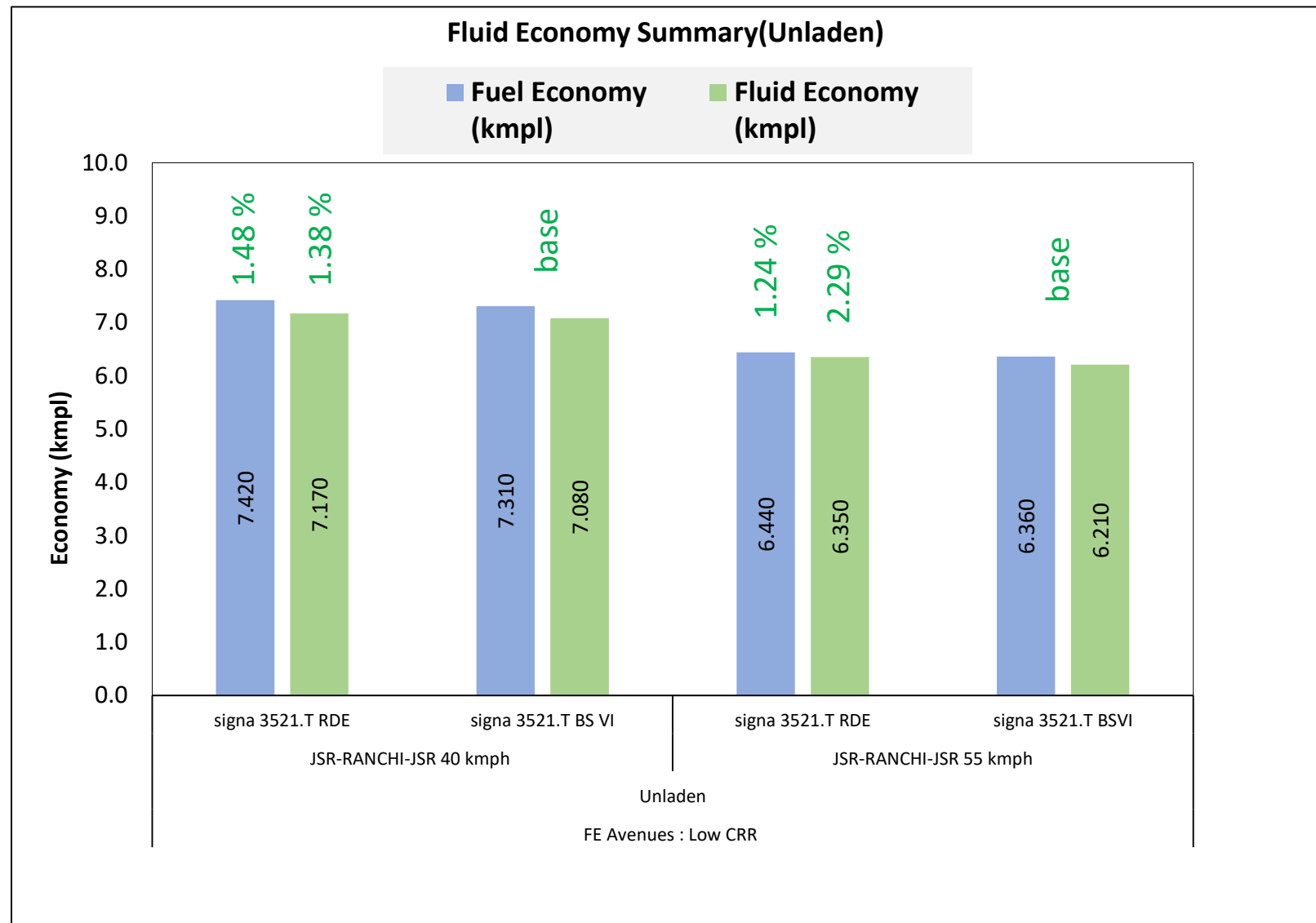


➤ Tractive effort of TML Signa 3521.T RDE 5.58 RAR vehicle is better than Signa 3521.T BSVI 5.12 RAR vehicle.

Summary FE Status with all FE Avenues(Laden)



Summary FE Status with all FE Avenues(Unladen)



Torque Table

Base vehicle (Signa 3521.T BS-VI OBD-I (200 HP))

FE Switch Torque Table (Torque Limit)										
	Load Gea	1st Gear	2nd Gear	3rd Gear	4th Gear	5th Gear	6th Gear	7th Gear	8th Gear	Reverse
Economy Mode	-	850	850	700	590	590	590	-	-	850
Balance Mode	-	850	850	700	700	700	700	-	-	850
Power Mode	-	850	850	850	850	850	850	-	-	850

Test vehicle (Signa 3521.T BS-VI OBD-II (204 HP))

FE Switch Torque Table (Torque Limit)										
	Load Gea	1st Gear	2nd Gear	3rd Gear	4th Gear	5th Gear	6th Gear	7th Gear	8th Gear	Reverse
Economy Mode	NA	850	850	700	590	590	590	-	-	850
Balance Mode	NA	850	850	700	700	700	700	-	-	850
Power Mode	NA	850	850	850	850	850	850	-	-	850

Duty Cycle Analysis Jamshedpur-Ranchi Road FE Trial

Duty Cycle : Summary (Laden)

JSR-RANCHI-JSR Duty Cycle

		26.04.2023 (40 kmph)	24.04.2023 (55 kmph)
Parameters		Signa 3521.T RDE Test Vehicle	Signa 3521.T RDE Test Vehicle
Engine idle speed (rpm)		750	750
Low idle time (%)		--	--
Average engine speed (rpm)		1218.3	1413.6
Average power (kW)		--	--
Avg. vehicle speed (kmph)	Including Stop	34.47	43.14
	Excluding Stop	--	--
FE switch utilization (%time)	Power	24.3	11.3
	Balance	23.0	6.1
	Eco	52.7	82.6
Gear utilization (%time)	Neutral Gear	6.5	4.2
	4 th Gear	4.1	3.2
	5 th Gear	12.4	6.8
	6 th Gear	68.8	79.7
	Total shifts	--	--
Throttle (%time)	100%	14.8	39.5
	0%	38.6	27.9
Coolant Temp (°C)	70-80°C	23.4	4.9
	80-90°C	37.6	33.7
	90-100°C	29.6	61.4
SCRTM Fuel spent (%)		--	--
Regeneration Fuel spent (%)		0.00	0.00
Duty cycle fuel economy (kmpl)		4.10	5.12
Duty cycle Distance(km)		227.0	222
Duty Cycle Time(Hrs)		--	--
Actual Avg. Fuel Economy		3.86	3.55

Duty Cycle : Summary (Unladen)

JSR-RANCHI-JSR Duty Cycle

		03.05.2023 (40 kmph)	06.05.2023 (55 kmph)
Parameters		Signa 3521.T RDE Test Vehicle	Signa 3521.T RDE Test Vehicle
Engine idle speed (rpm)		750	750
Low idle time (%)		--	--
Average engine speed (rpm)		1052.3	1420.2
Average power (kW)		--	--
Avg. vehicle speed (kmph)	Including Stop	32.64	43.07
	Excluding Stop	--	--
FE switch utilization (%time)	Power	2.2	0.4
	Balance	1.5	0.8
	Eco	96.3	98.8
Gear utilization (%time)	Neutral Gear	7.4	10.4
	4 th Gear	1.1	1.3
	5 th Gear	1.6	2.1
	6 th Gear	86.8	82.1
	Total shifts	--	--
Throttle (%time)	100%	0.1	2.1
	0%	30.2	29.7
Coolant Temp (°C)	70-80°C	34	26.3
	80-90°C	1.5	60.8
	90-100°C	0.2	9.2
SCRTM Fuel spent (%)		--	--
Regeneration Fuel spent (%)		0.00	0.00
Duty cycle fuel economy (kmpl)		7.14	6.32
Duty cycle Distance(km)		225.0	225
Duty Cycle Time(Hrs)		--	--
Actual Avg. Fuel Economy		7.10	6.50

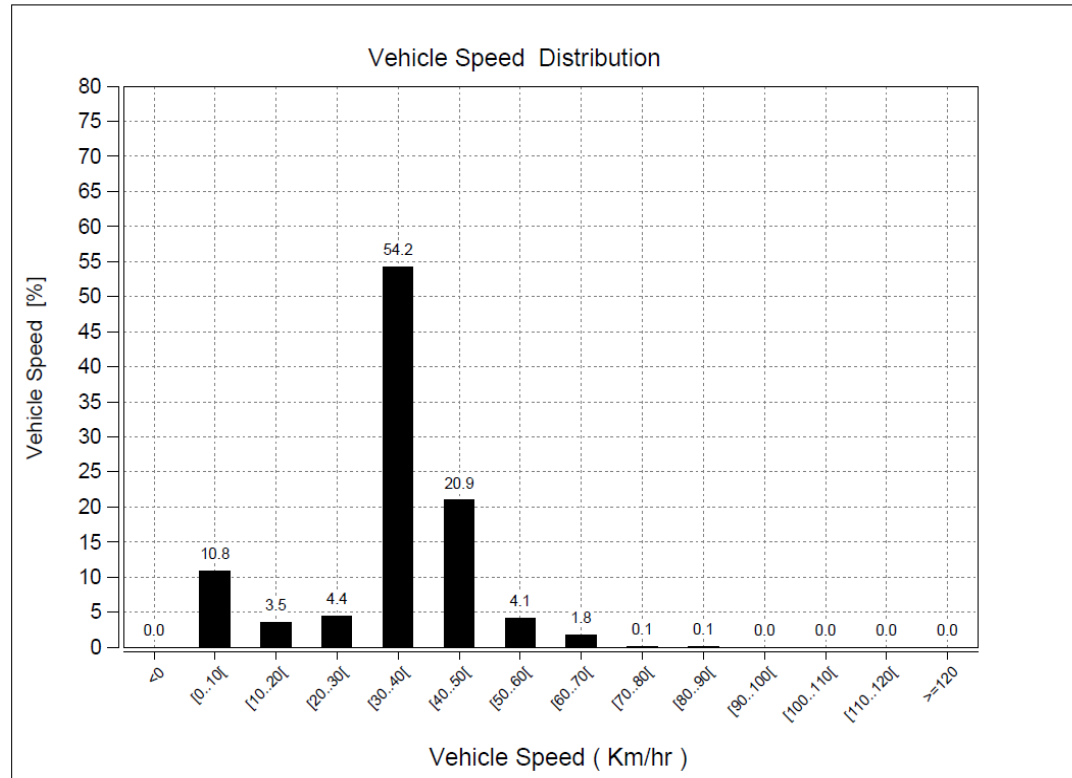
Vehicle Speed Distribution

Laden Trial

Signa 3521.T RDE Test Vehicle

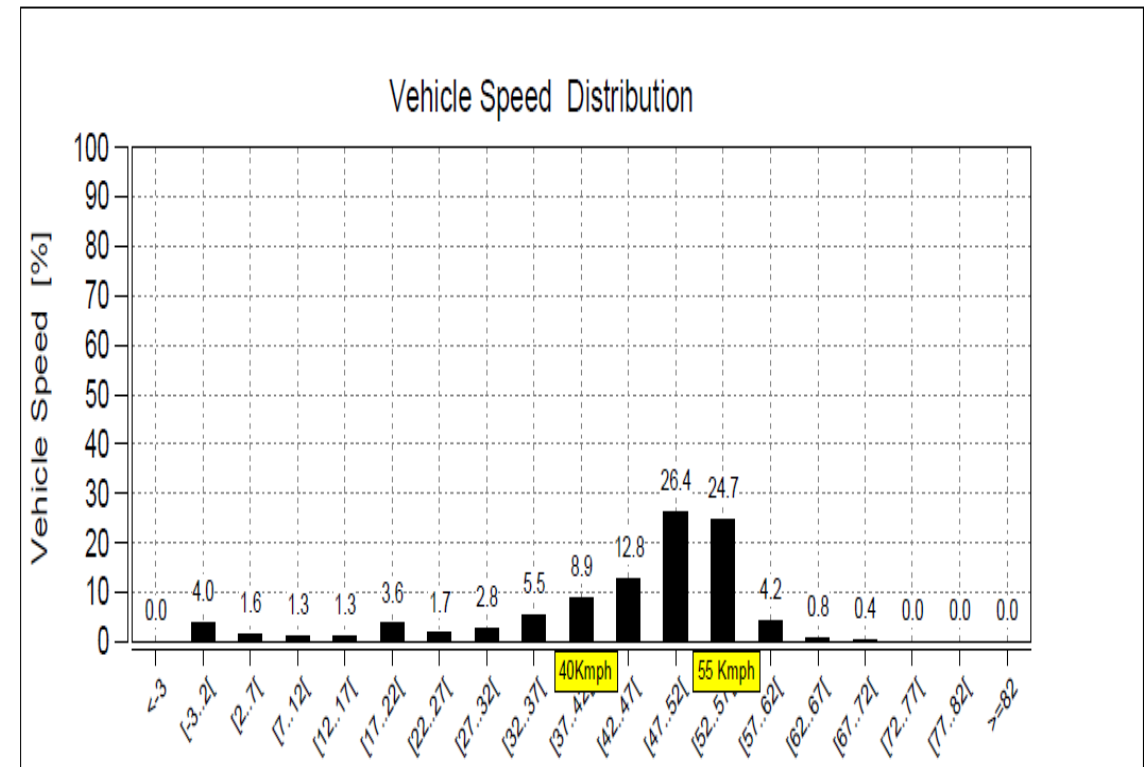
26.04.23

40 kmph



24.04.23

55 kmph



Vehicle Speed Distribution

Unladen Trial

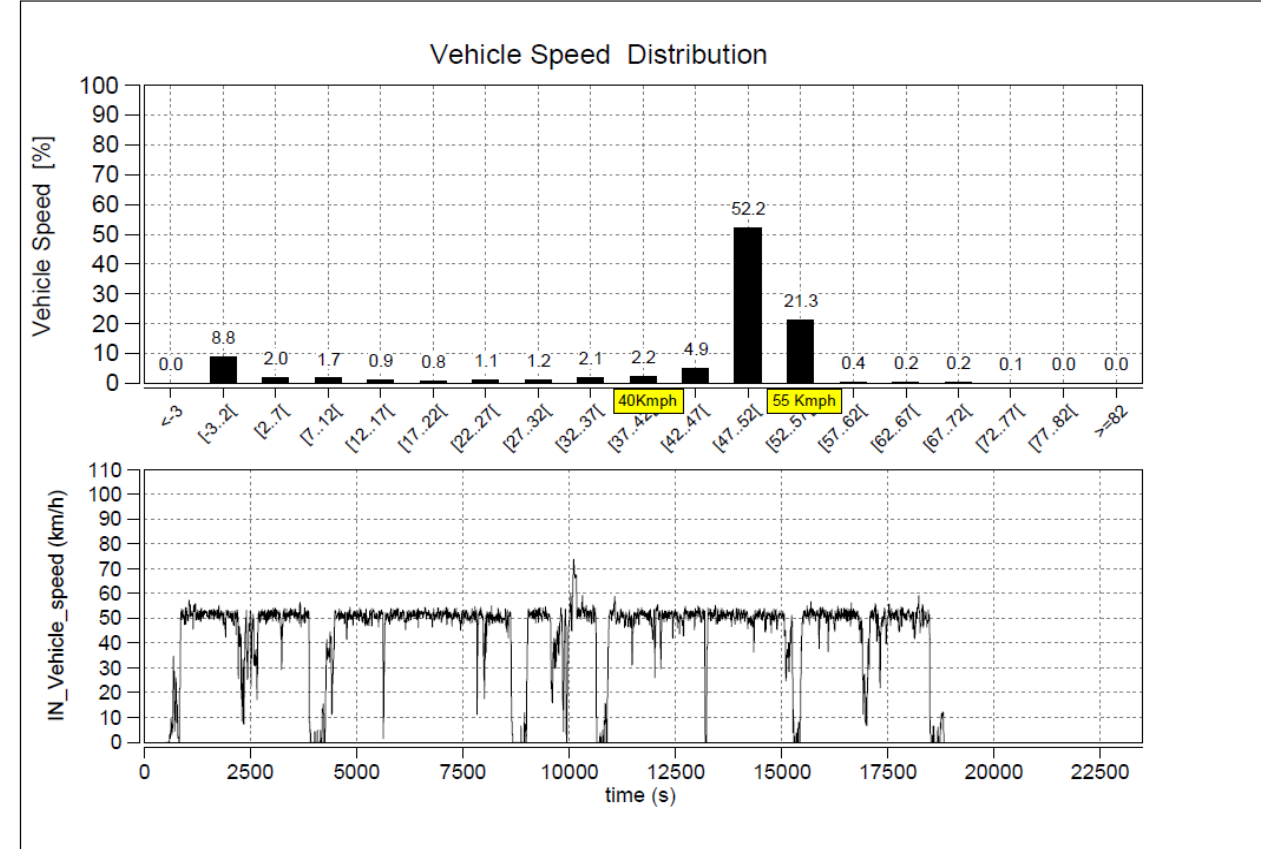
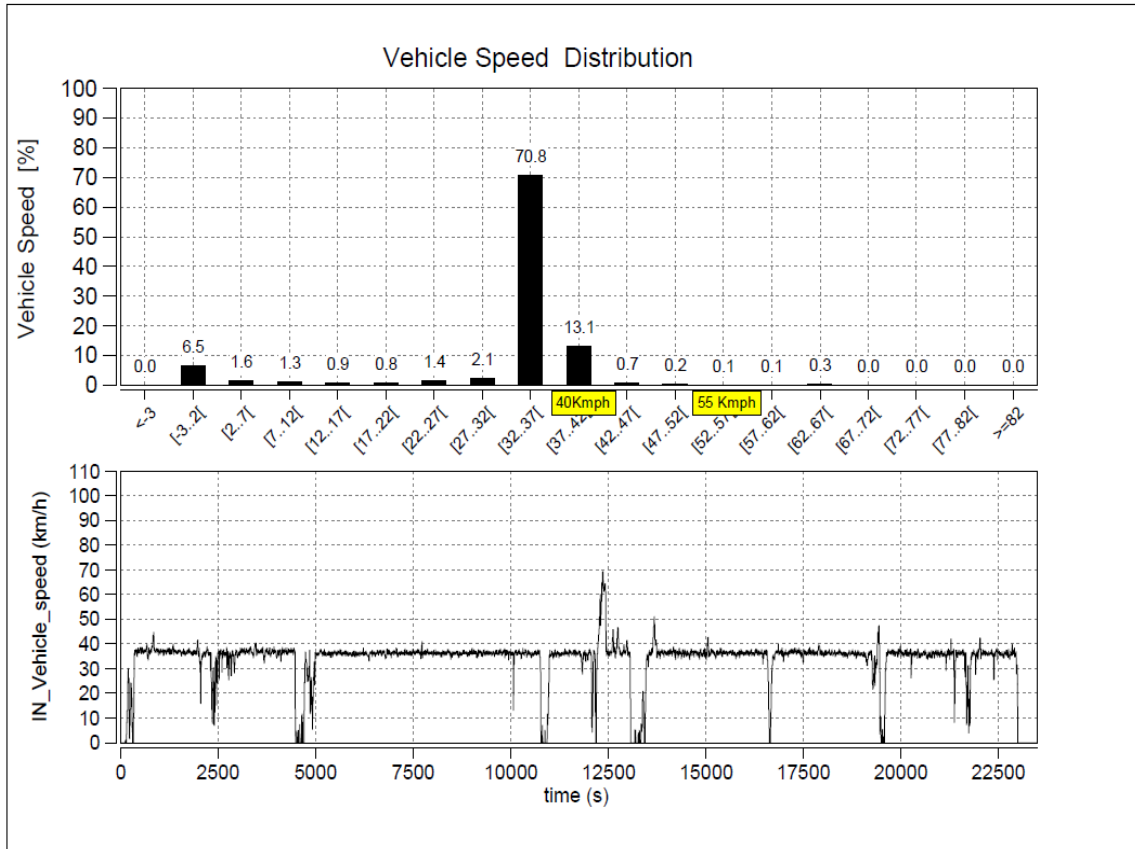
Signa 3521.T RDE Test Vehicle

03.05.23

40 kmph

06.05.23

55 kmph



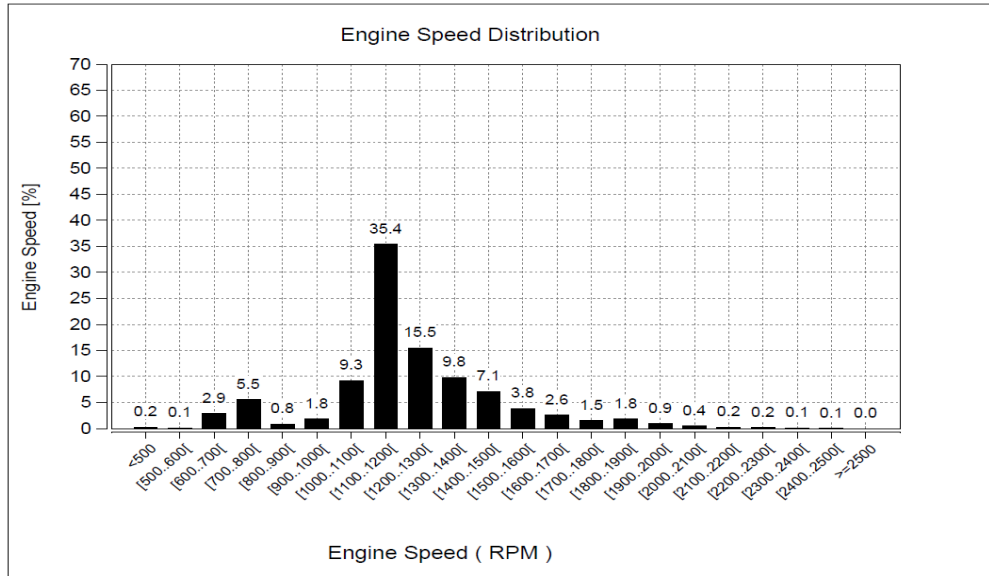
Engine Speed Distribution

Laden Trial

Signa 3521.T RDE Test Vehicle

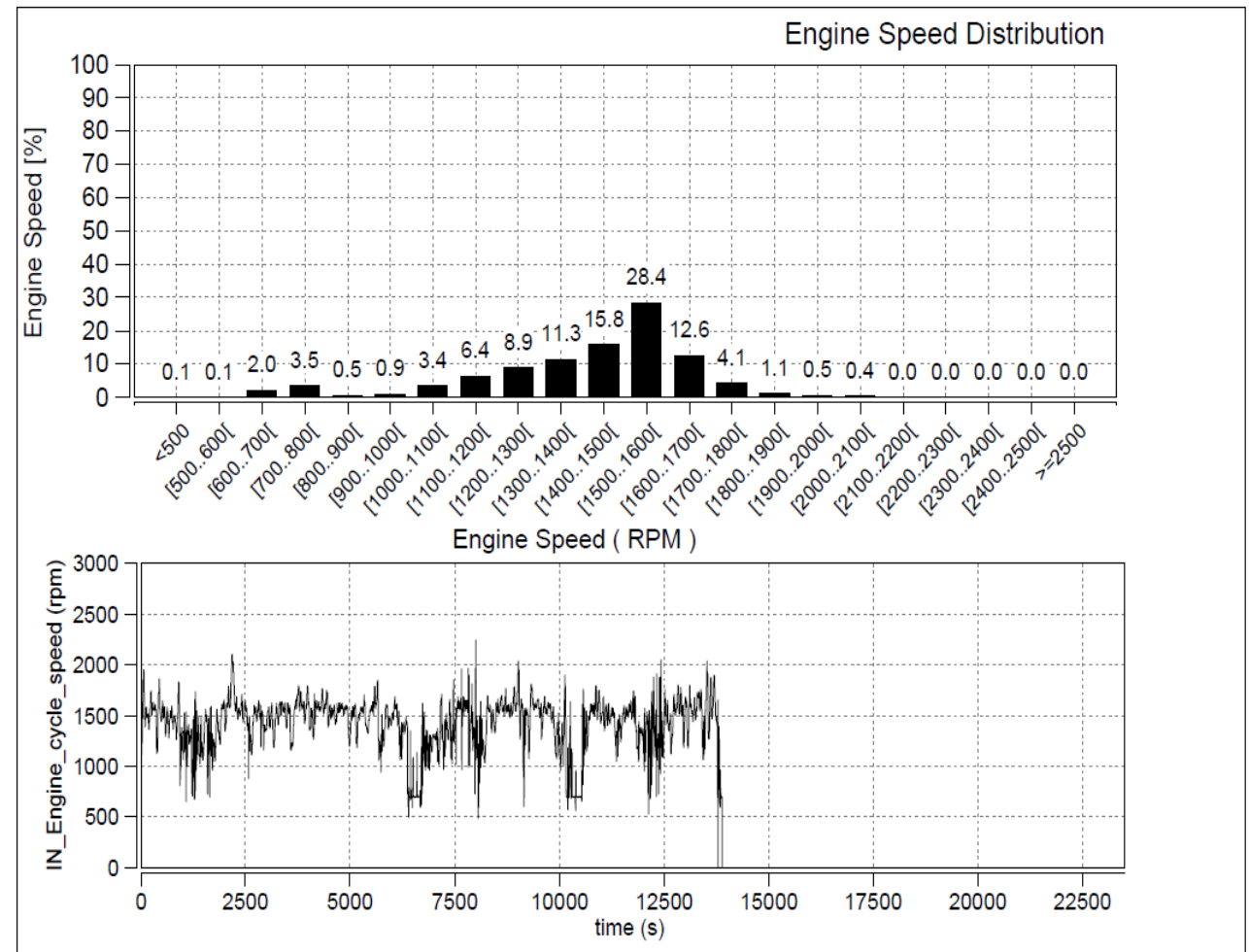
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40 kmph



24.04.23

55 kmph



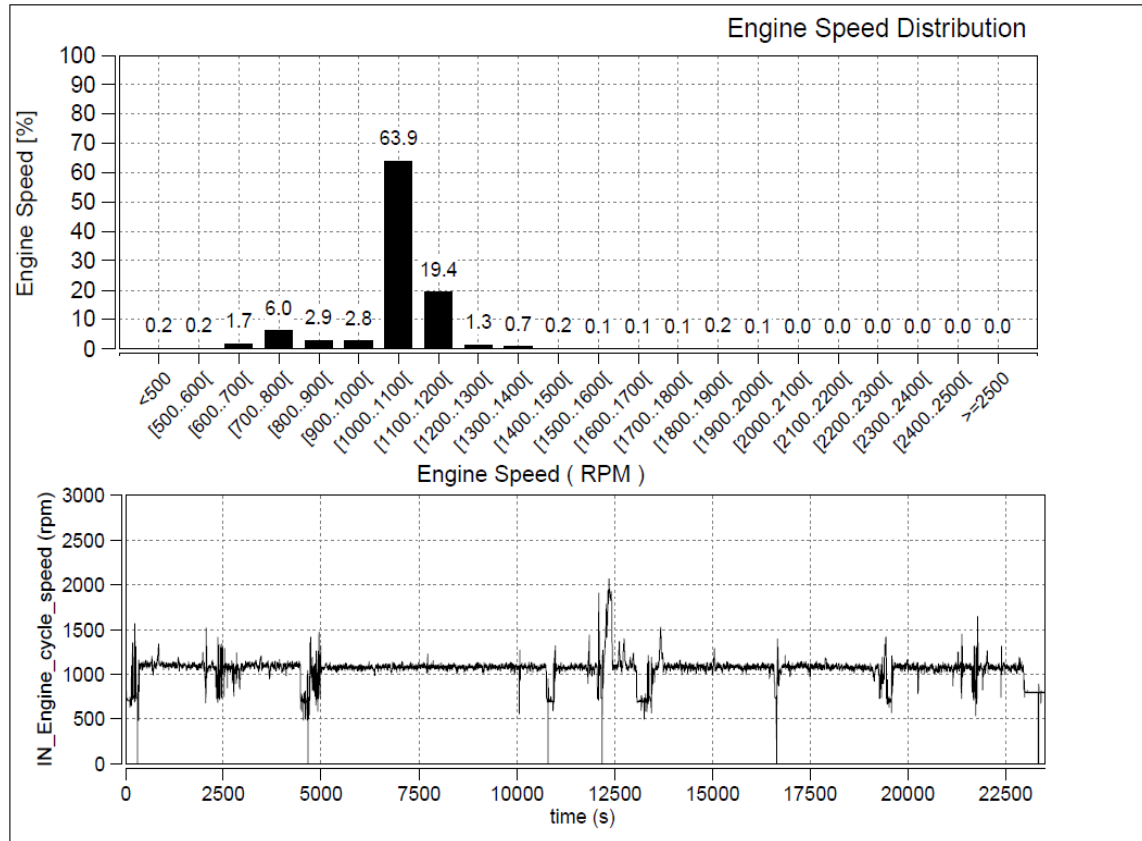
Engine Speed Distribution

Unladen Trial

Signa 3521.T RDE Test Vehicle

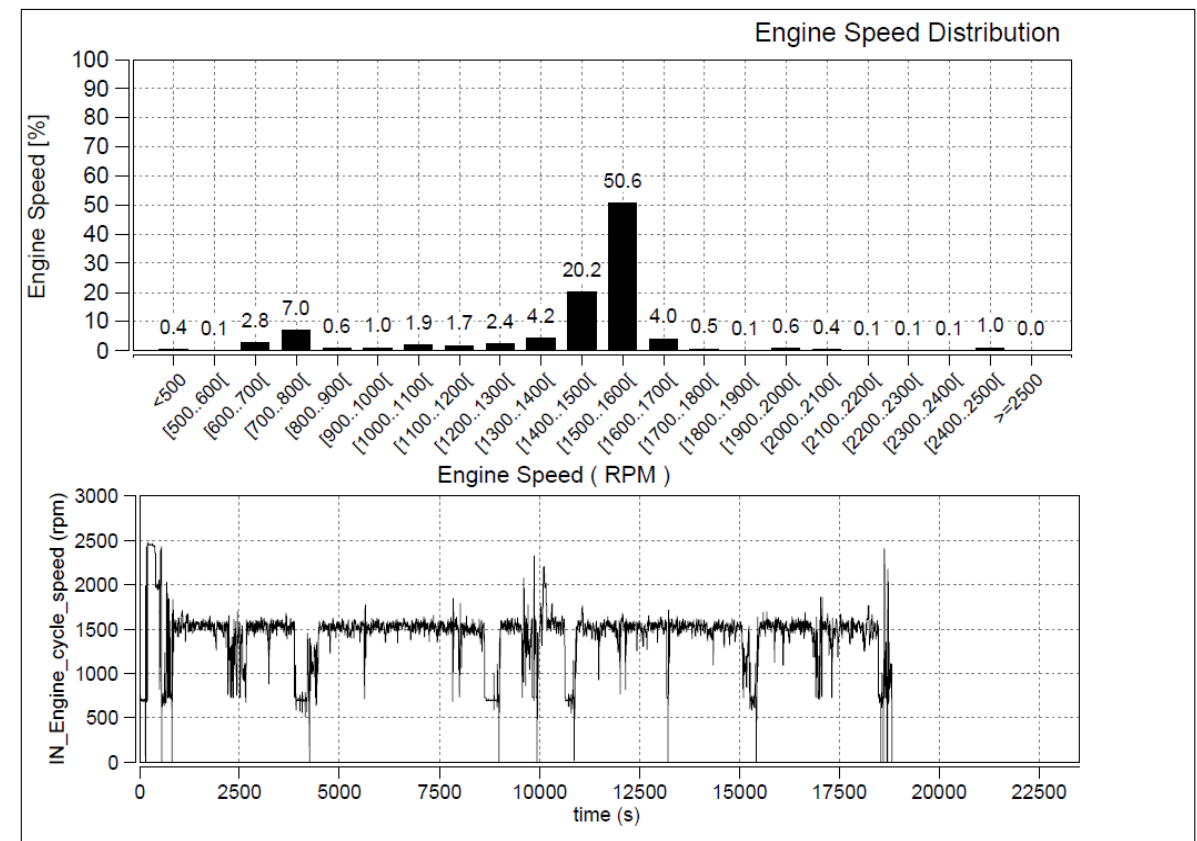
03.05.23

40 kmph



06.05.23

55 kmph



Laden Trial

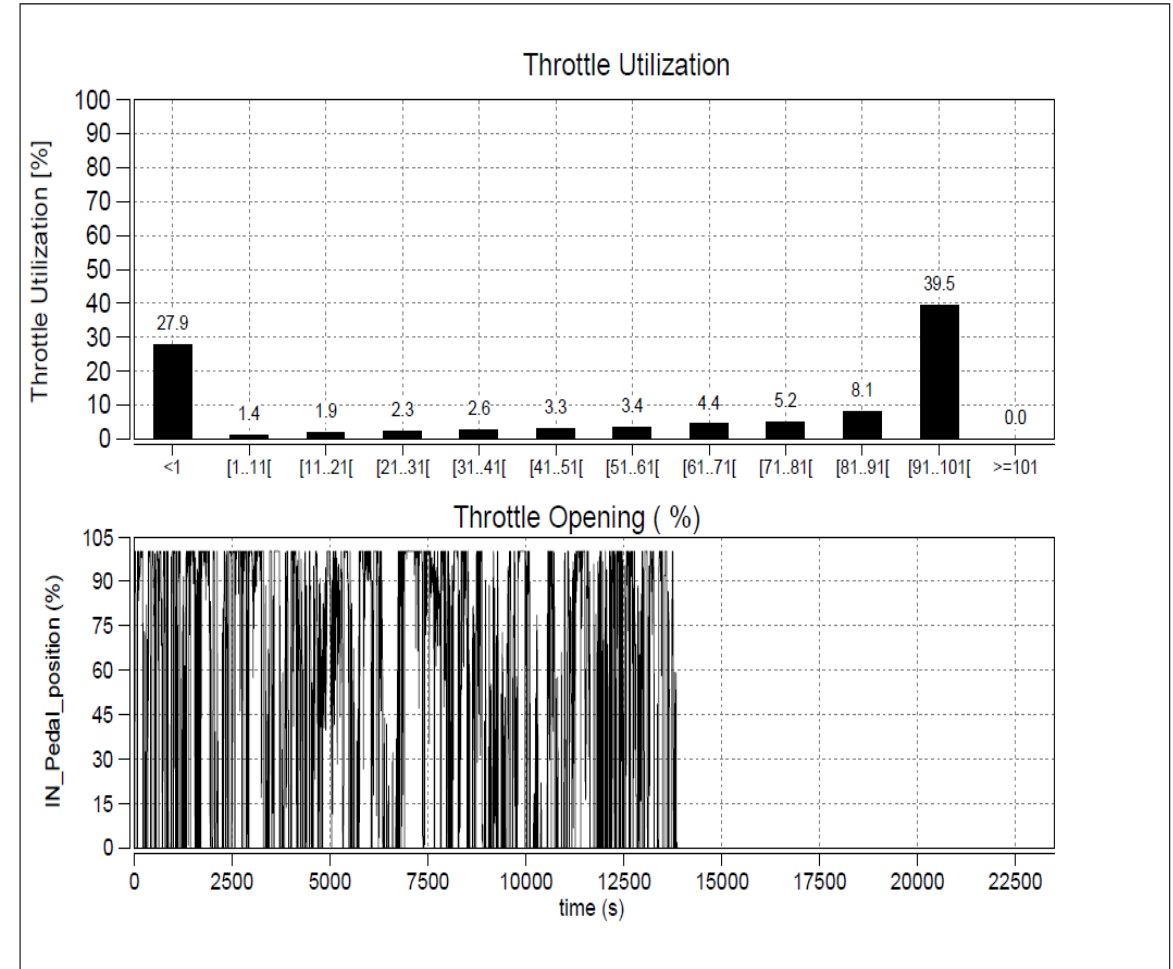
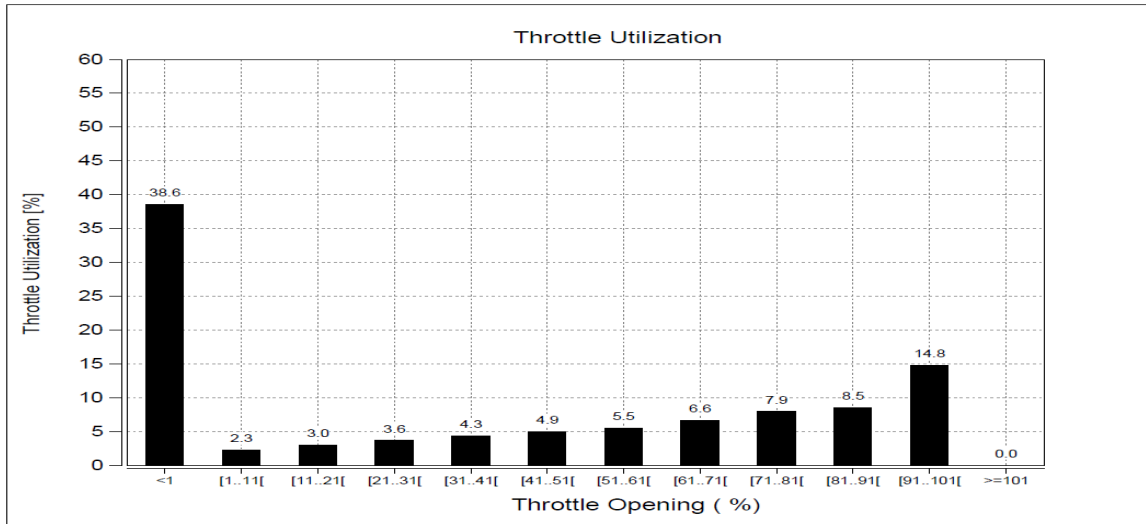
Signa 3521.T RDE Test Vehicle

26.04.23

40 kmph

24.04.23

55 kmph



Unladen Trial

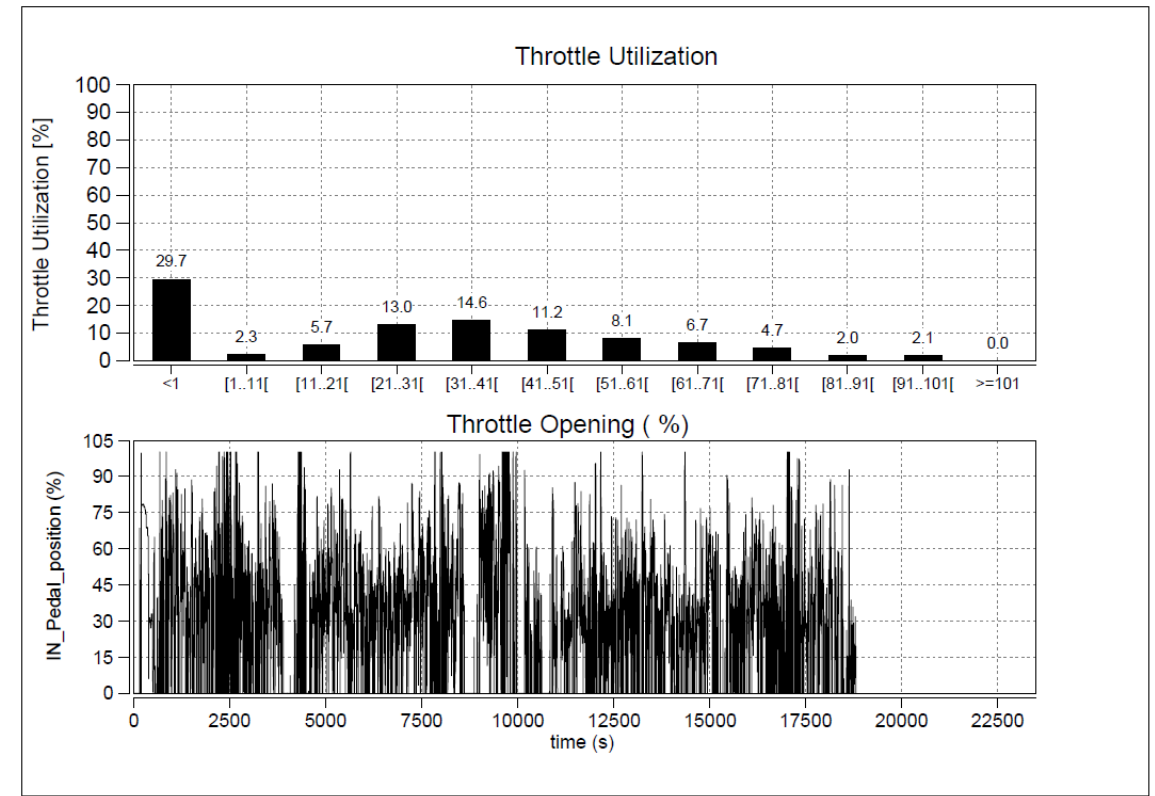
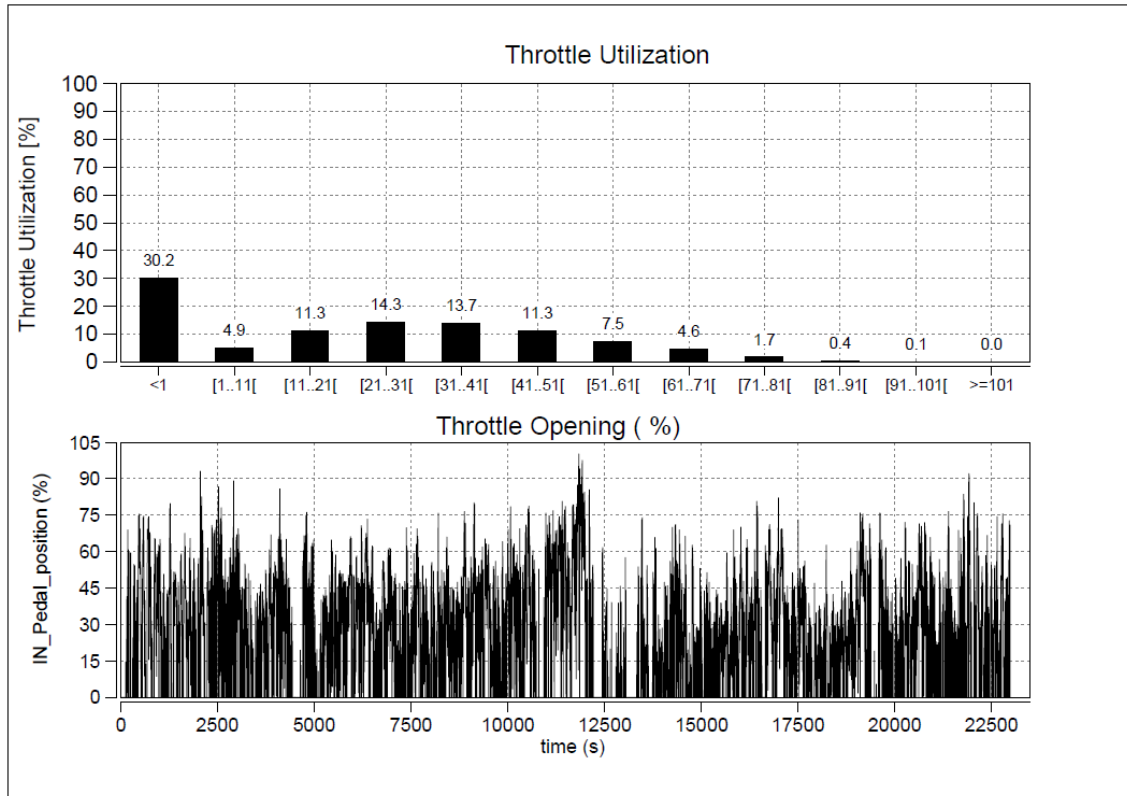
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03.05.23

40 kmph

06.05.23

55 kmph

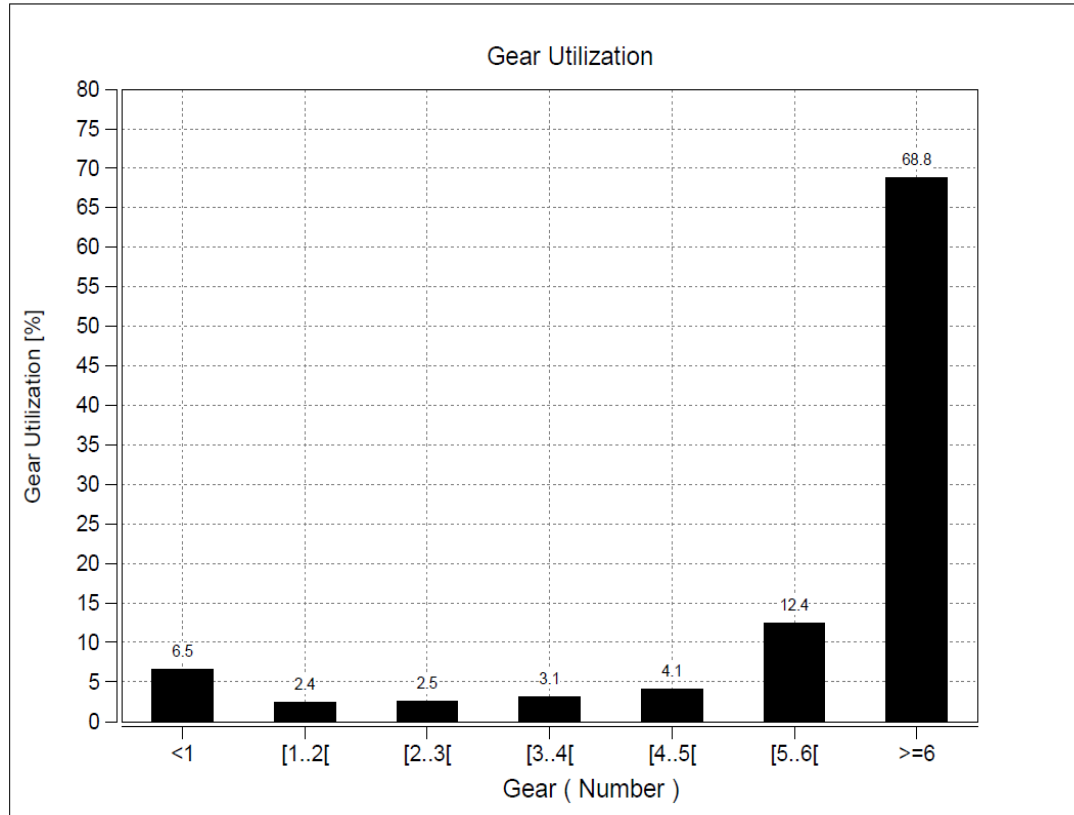


Laden Trial

Signa 3521.T RDE Test Vehicle

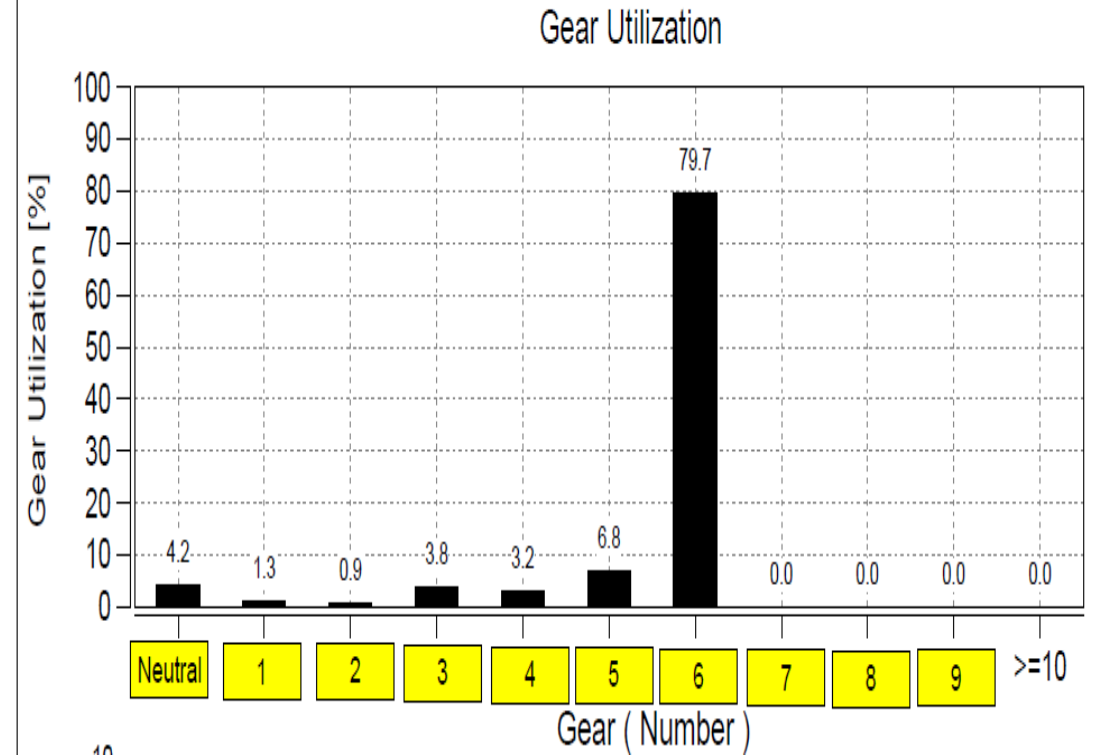
26.04.23

40 kmph



24.04.23

55 kmph



Unladen Trial

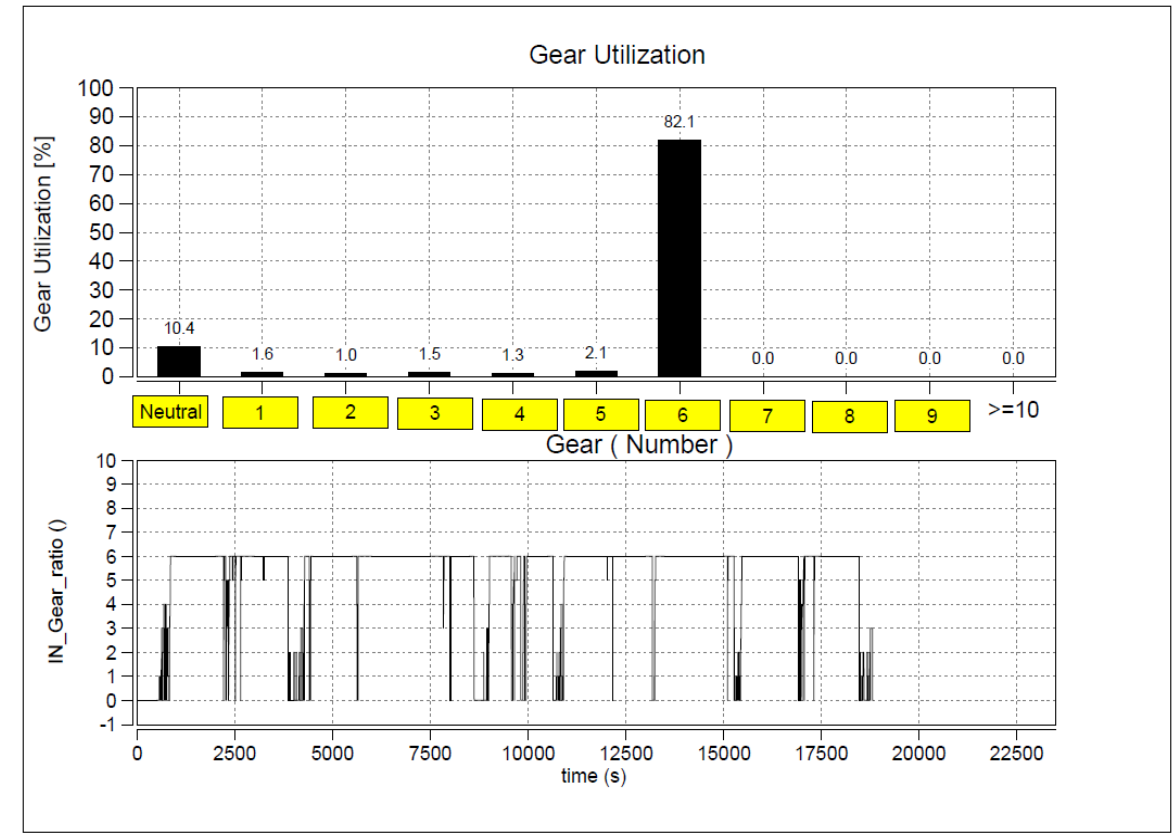
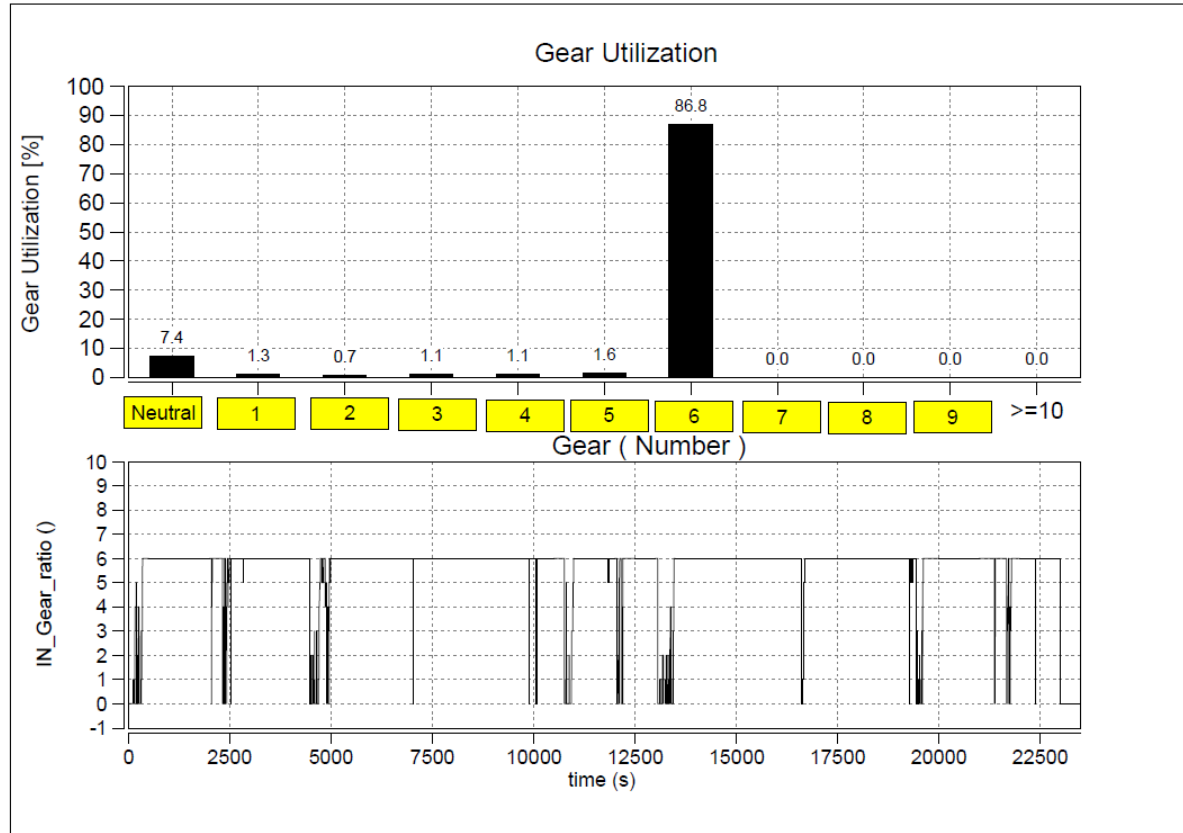
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03.05.23

40 kmph

06.05.23

55 kmph



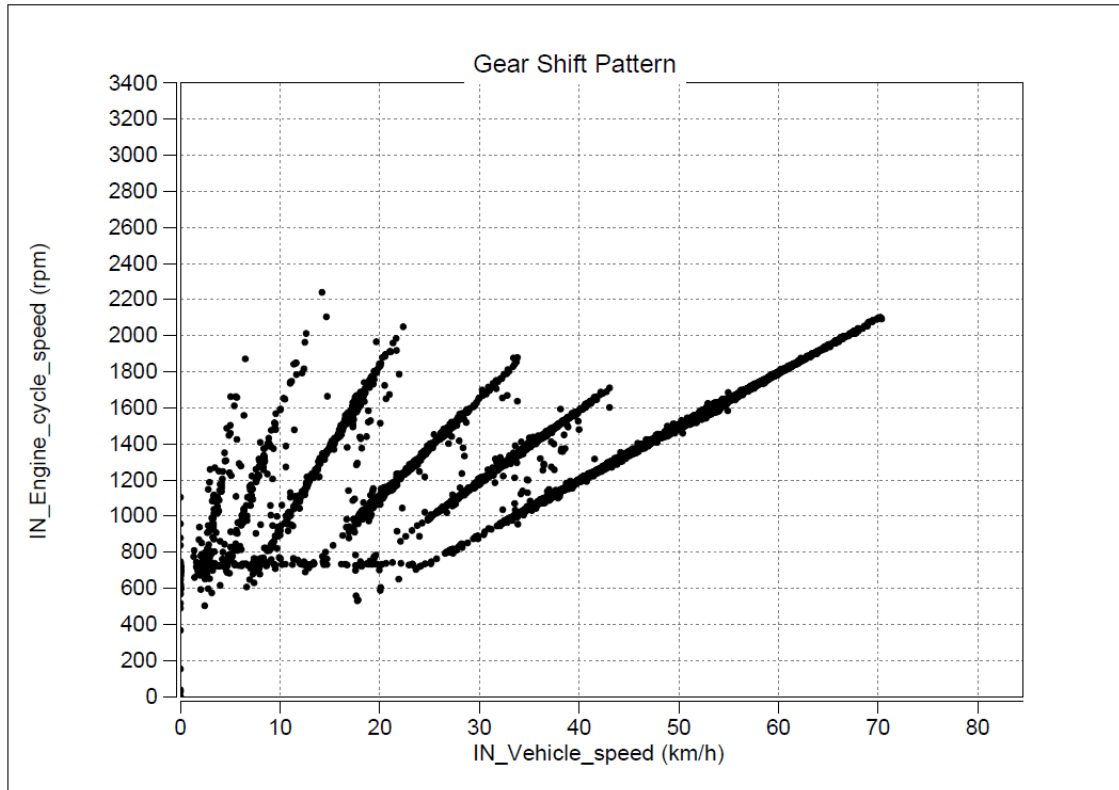
Transmission Shift Pattern

Laden Trial

Signa 3521.T RDE Test Vehicle

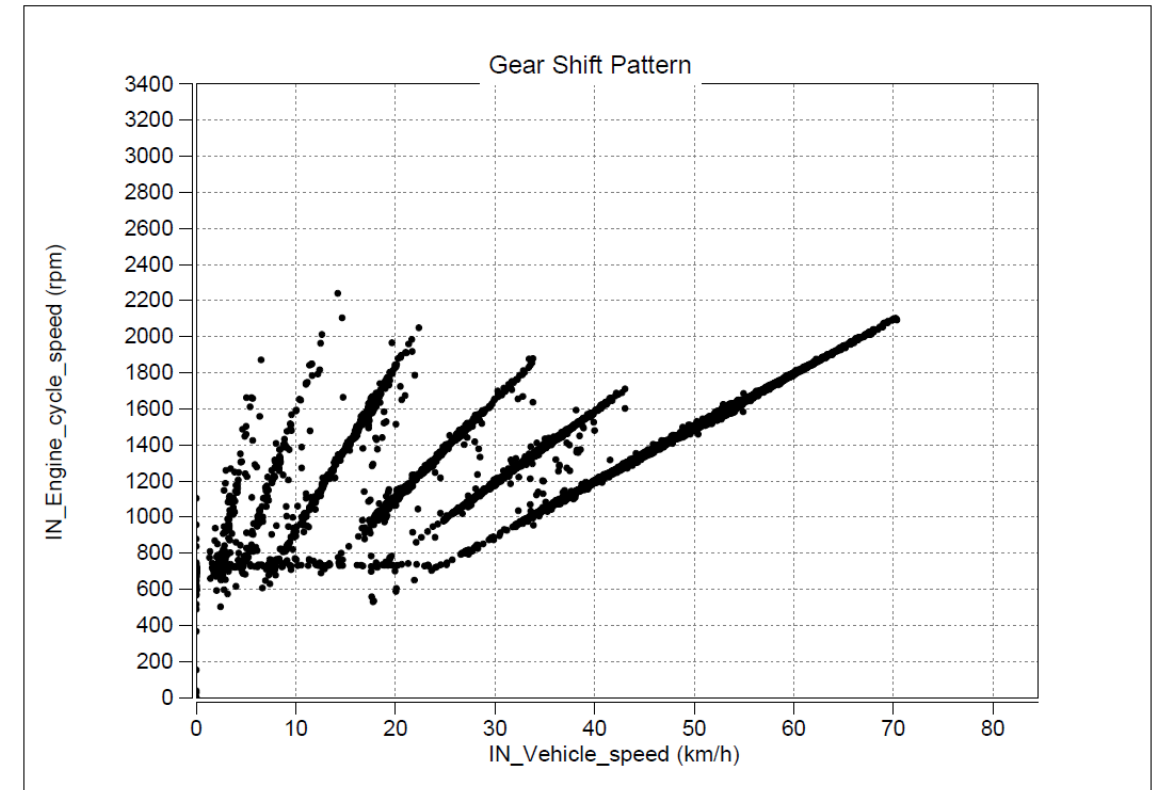
26.04.23

40 kmph



24.04.23

55 kmph



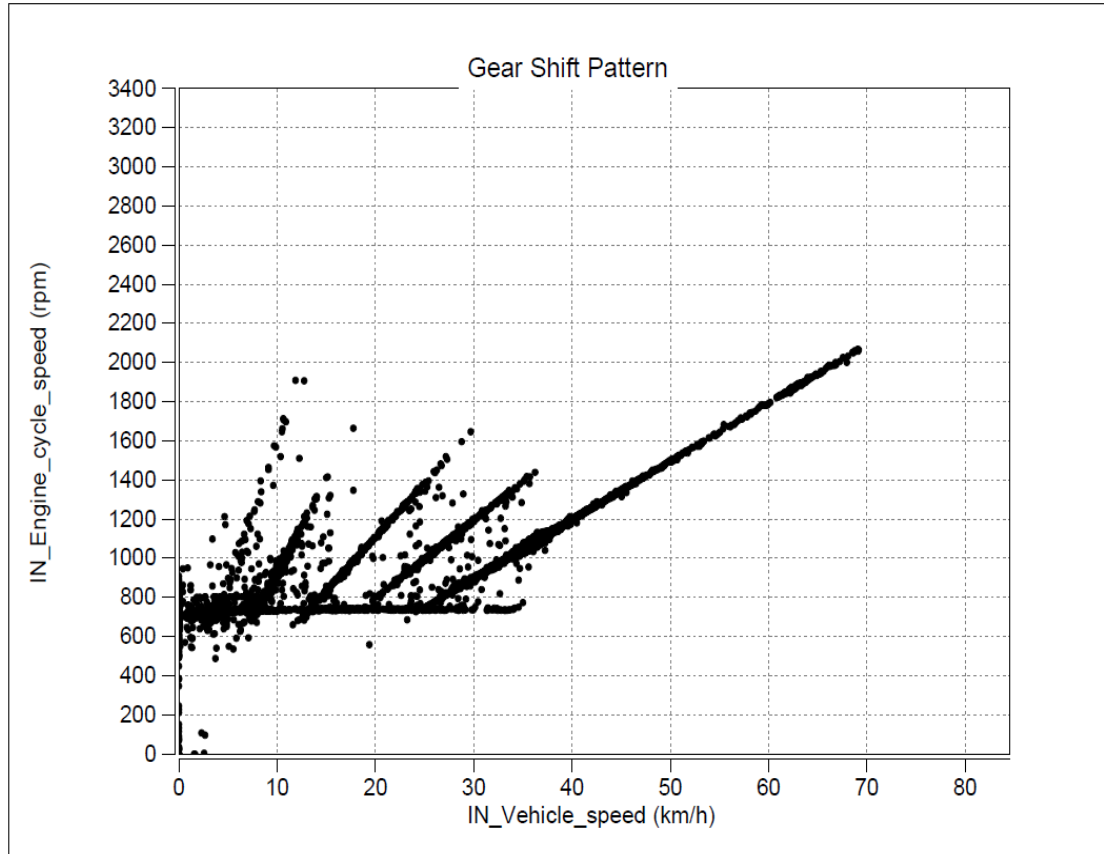
Transmission Shift Pattern

Unladen Trial

Signa 3521.T RDE Test Vehicle

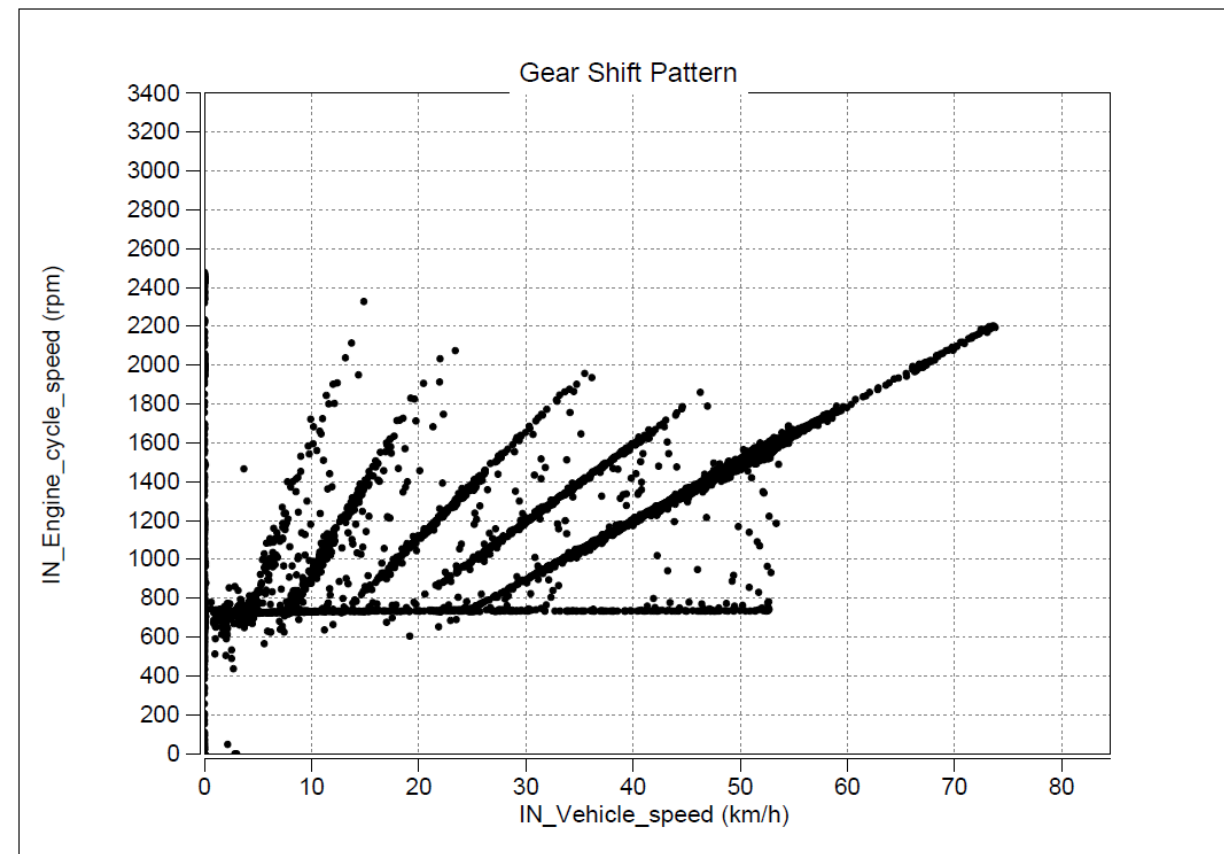
03.05.23

40 kmph



06.05.23

55 kmph



Laden Trial

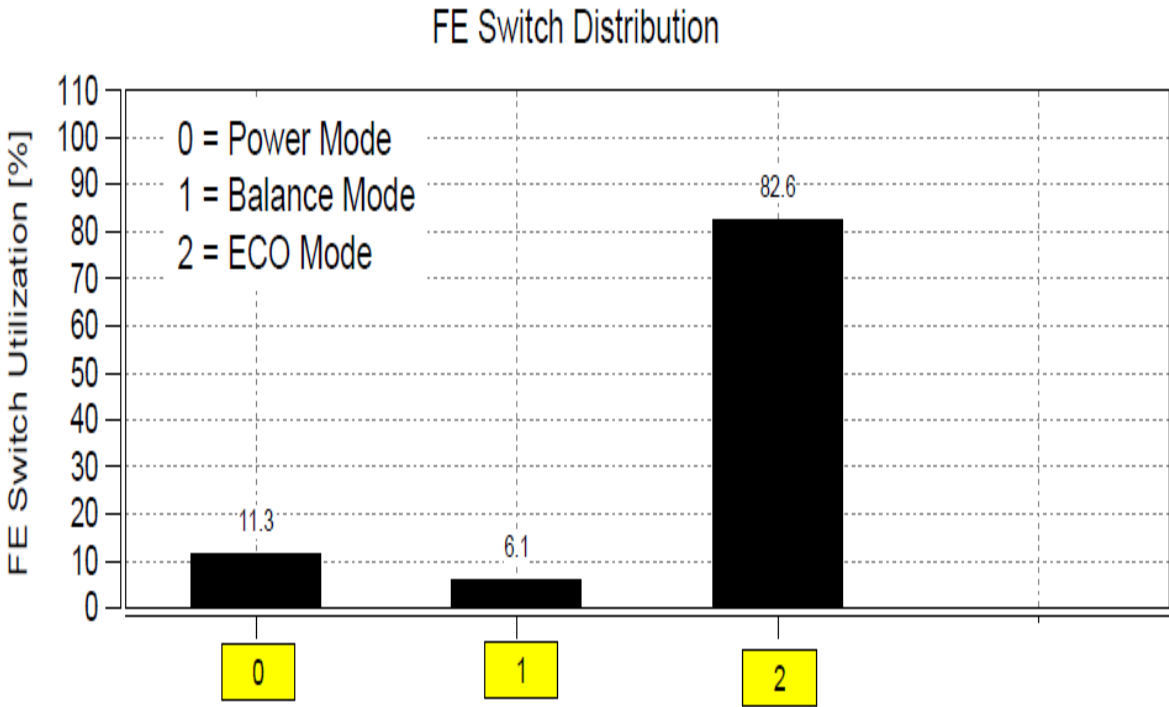
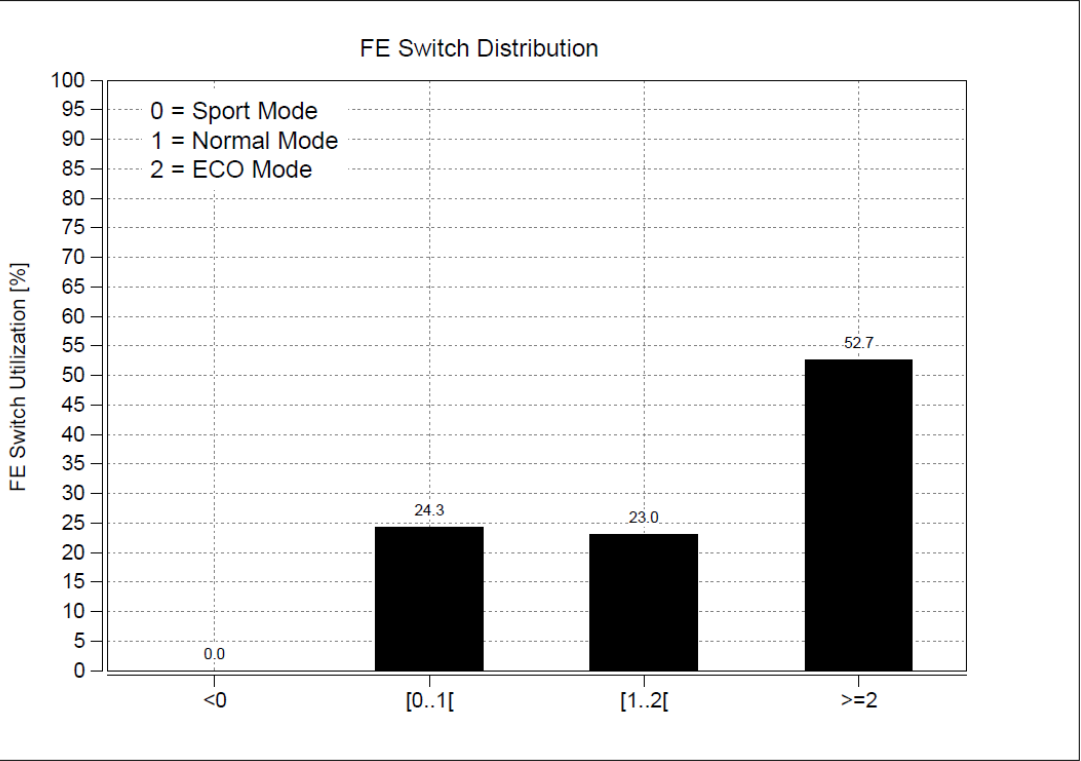
Signa 3521.T RDE Test Vehicle

26.04.23

40 kmph

24.04.23

55 kmph



Unladen Trial

Signa 3521.T RDE Test Vehicle

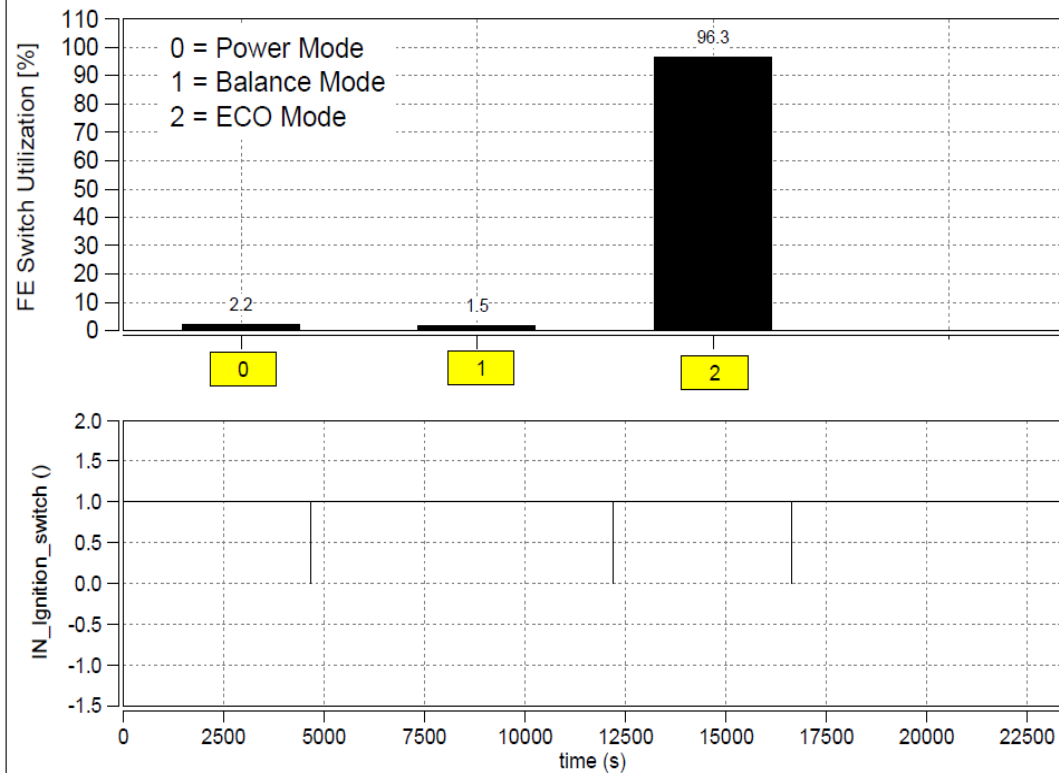
03.05.23

40 kmph

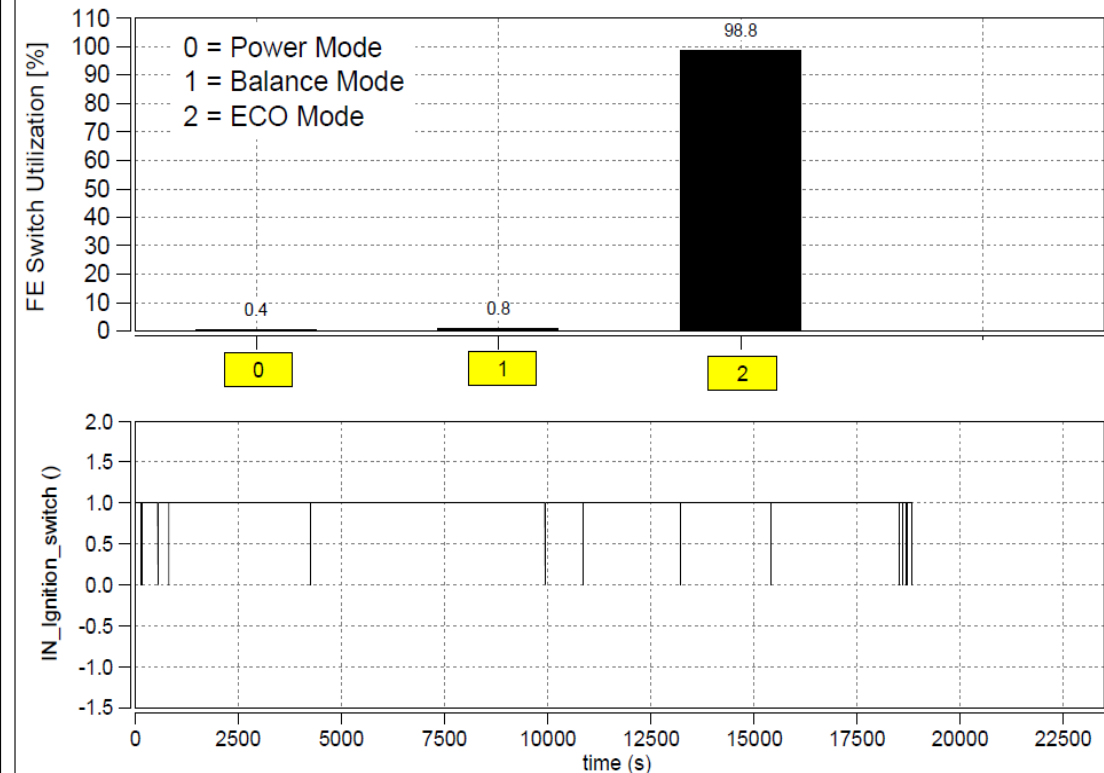
06.05.23

55 kmph

FE Switch Distribution



FE Switch Distribution



Coolant Temp Distribution

Laden Trial

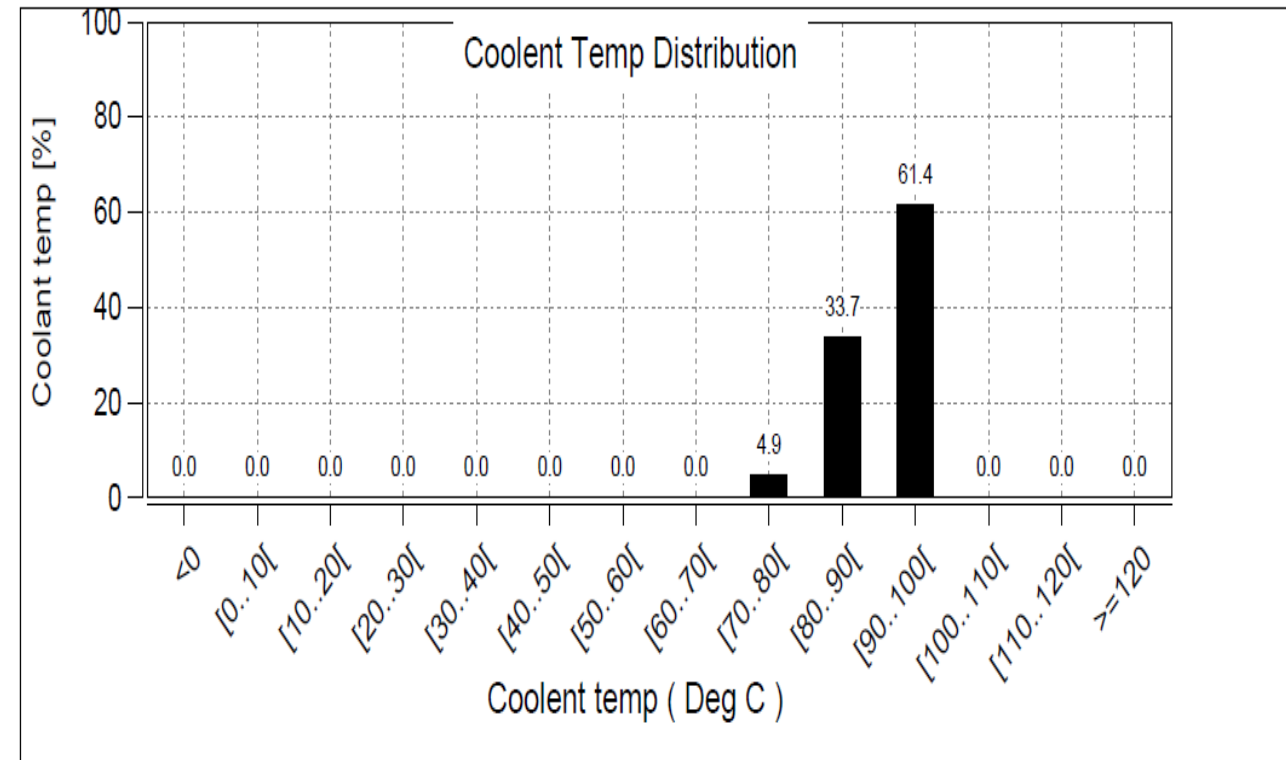
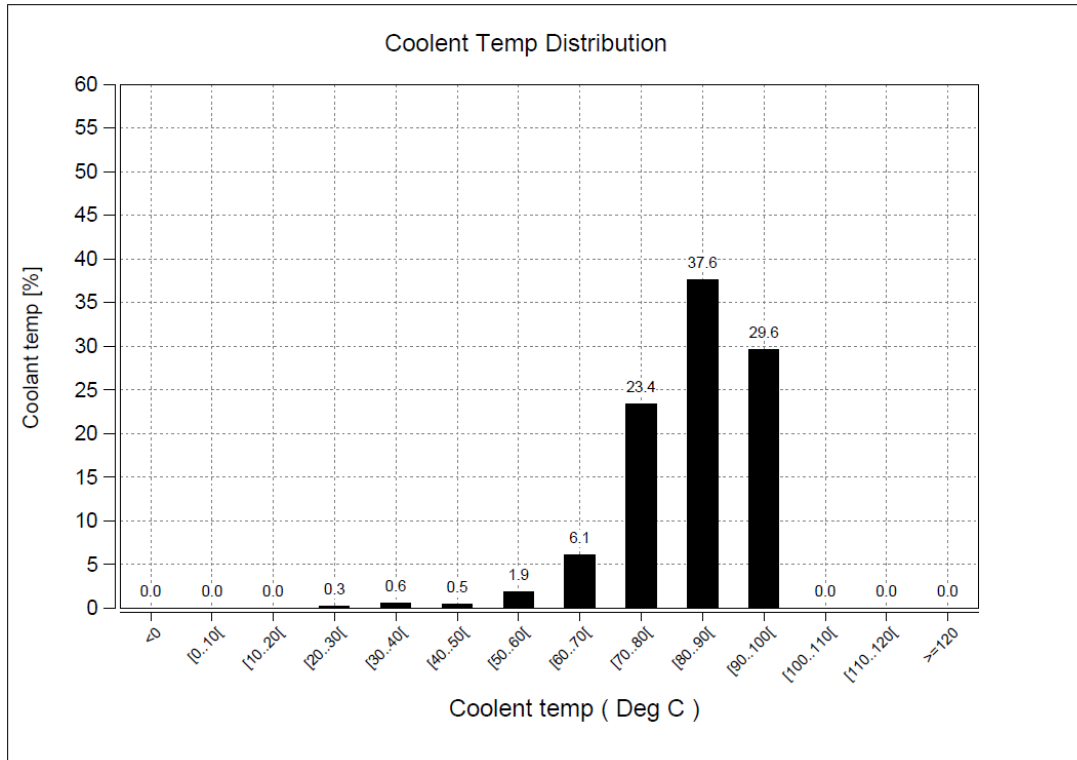
Signa 3521.T RDE Test Vehicle

26.04.23

40 kmph

24.04.23

55 kmph



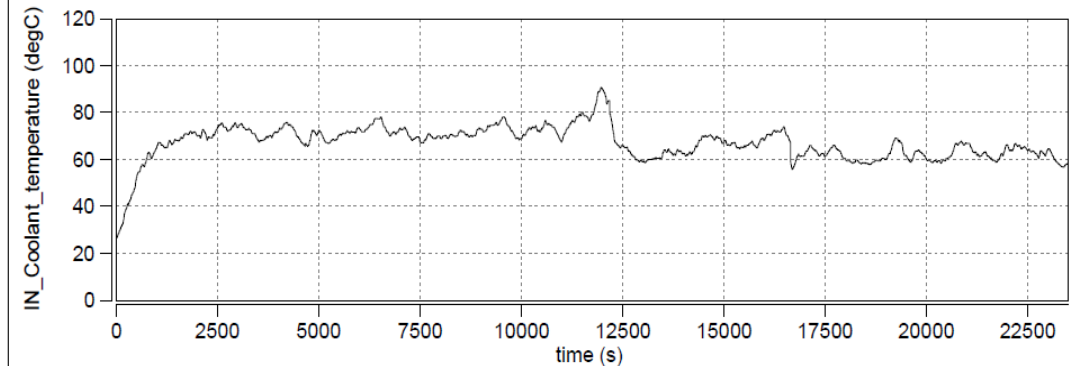
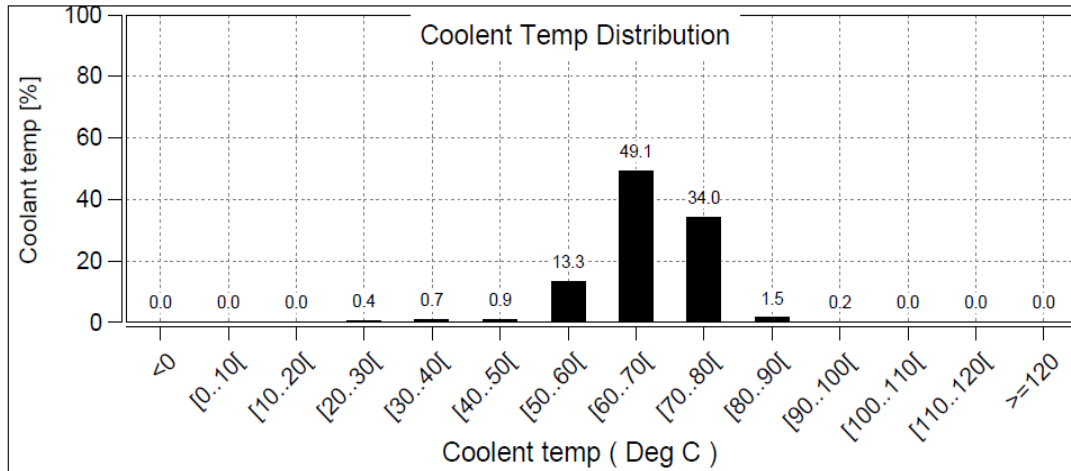
Coolant Temp Distribution

Unladen Trial

Signa 3521.T RDE Test Vehicle

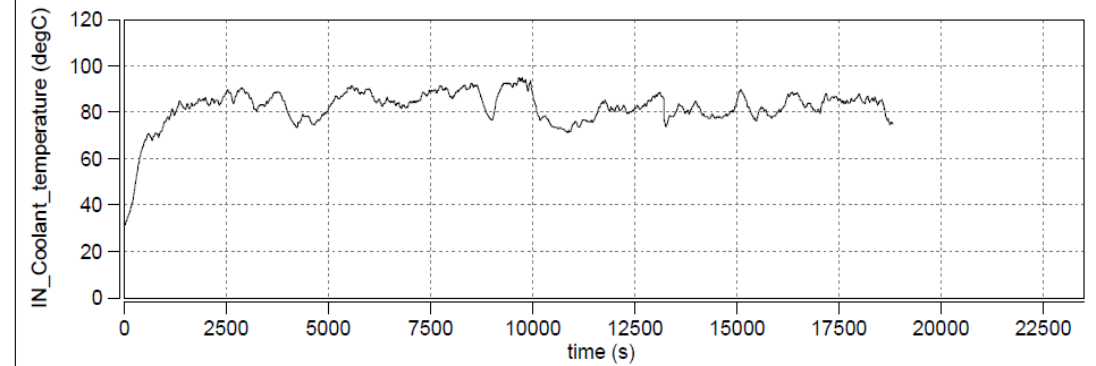
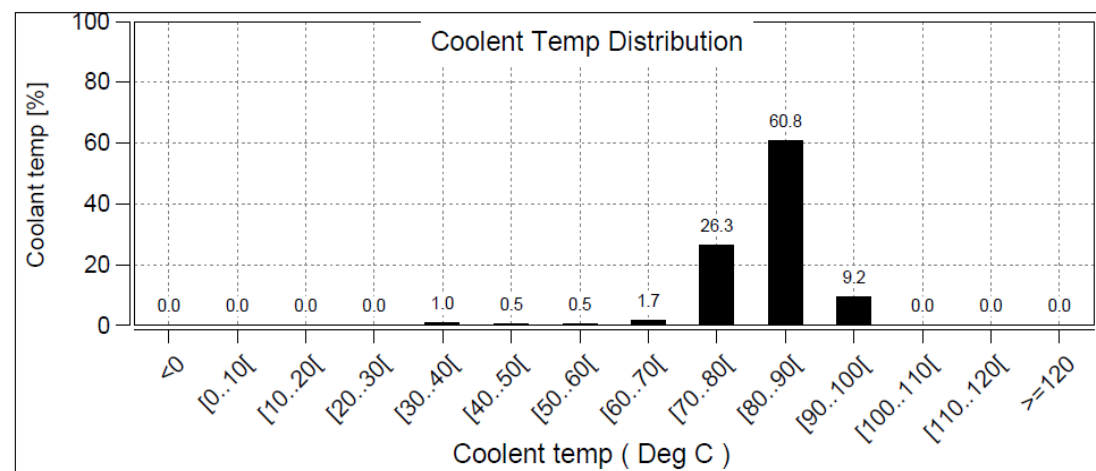
03.05.23

40 kmph



06.05.23

55 kmph



Fluid Economy(FLE) Status of 3521.T RDE Test vehicle with base vehicle 3521.T :

(THU contribution is deducted by 0.5 % from Test FLE value)

1. Laden

- a) 40kmph :- Fluid Economy of 3521.T is **superior** than base vehicle by 2.22 %.
- b) 55kmph :- Fluid Economy of 3521.T is **superior** than base vehicle by 3.84 %.

2. Unladen

- a) 40kmph :- Fluid Economy of 3521.T is **superior** than base vehicle by 1.38 %.
- b) 55kmph :- Fluid Economy of 3521.T is **superior** than base vehicle by 2.29 %.

3. (80% Laden + 20 % unladen)

- a) **40kmph** :- Fluid Economy of 3521.T is **superior** than base vehicle by 2.12 %.
- b) **55kmph** :- Fluid Economy of 3521.T is **superior** than base vehicle by 3.65 %.

Thank You

TEST REQUISITION

Ref.	NA	Date:	31/03/2023
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To,

ERC Testing

Kindly arrange to take up for testing of the following component.

1	Part Description	TATA SIGNA 3521.T 5L BSVI				
2	Part Number	51705156R,				
3	Drawing number	517000000003,	Mod status	B,2,	Mod Date	10/07/2023
4	Vehicle Model / Project	TATA SIGNA 3521.T RDe BSVI				
5	WBS Number	P.22.EA.501 -RDE Truck 5.6 Ltr 3123.T / 3523.T				
6	Supplier	NA				
7	Number of samples					
8	Inspection report	Enclosed	☒ Yes	☐ No	If No commitment date:	
9	Material report	Enclosed	☐ Yes	☒ No	If No commitment date:	
10	Suppliers test report	Enclosed	☒ Yes	☐ No	If No commitment date:	
11	Reason for testing	First Source Development	☐	Weight Reduction	☐	
		Alternate source development	☐	Cost Reduction	☐	
		Data generation Test	☐	Regulatory test	☐	
		Any other (pl. specify)	☐	New Development	☐	
12	What testing required?	DYNAMIC				
13	Remarks					
14	Peripheral Parts Availability	YES				

Authorized Signature:	
Name:	
Dept:	

Contact Person:	
Phone Number	

For Testing Dept Use only	Test Item Number and Date(stamp)	Test Engineer

TATA MOTORS LIMITED ERC PUNE	TEST PLAN		DATE: 13/07/2023			
	VEHICLE TESTING DEPARTMENT					
Description: TATA SIGNA/LPT 3521.T BSVI			Test Item No: 42503			
			New	✓	Moderate	✗
			Major	✗	Minor	✗

Part No: 51705156R,	Modification No: B,2,	Enclosures: Nil
Drawing No: 5170000000003,		
Make: NA	WBS Number: P.22.EA.501 -RDE Truck 5.6 Ltr 3123.T / 3523.T	Copies to :
Vehicle Type: TATA SIGNA 3521.T RDe BSVI	No. of Samples: 1	
Test Requested by:	VEHICLE-I	Equipments: Nil
Test Requisition Reference:	Testing-Outdoor/15056	
Reason for Test/s:	Component & System Level Performance Tests	
Regulatory Requirement:	a) Component: NA	
	b) Vehicle: NA	

Sr. No.	Test description	Test specification	Acceptance criteria	No. of Samples	Test Responsibility	Proto Phase (Mule/ Alpha/ Beta/PP)	Remarks
1	Back to back FE of Signa 3523.T	As per PED	As per PALS	1	Debabrata Das	Beta	Completed

REMARKS: Test plan for DVP tests	INTERNAL REF:	
TEST TO FAILURE: No		
MEOST: No		
RELIABILITY TESTING: No		
BENCHMARKING TESTING: No		
DFMEA/FMEA DONE: No		
PREPARED BY	CHECKED BY	APPROVED BY